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RESEARCH ARTICLE

Deep Learning-Based Coding Strategy for Improved Cochlear Implant Speech Perception in Noisy Environments

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ABSTRACT Automatic speech recognition (ASR) and speech enhancement are essential tools in modern life, aiding not only in machine interaction but also in supporting individuals with hearing impairments. These processes begin with capturing speech in analog form and applying signal processing algorithms to ensure compatibility with devices like cochlear implants (CIs). However, CIs, with their limited number of electrodes, often cause speech distortion, and despite advancements in state-of-the-art signal processing techniques, challenges persist, particularly in noisy environments with multiple speech sources. The rise of artificial intelligence (AI) has introduced innovative strategies to address these limitations. This paper presents a novel deep learning (DL)-based technique that leverages attention mechanisms to improve speech intelligibility through noise suppression. The proposed approach includes two strategies: the first integrates temporal convolutional networks (TCNs) and multi-head attention (MHA) layers to capture both local and global dependencies within the speech signal, enabling precise noise filtering and improved clarity. The second strategy builds on this framework by additionally incorporating bidirectional gated recurrent units (Bi-GRU) alongside TCN and MHA layers, further refining sequence modeling and enhancing noise reduction. The optimal model configuration, using TCN-MHA-Bi-GRU with a kernel size of 16, achieved a compact model size of 788K parameters and recorded training, and validation losses of 0.0350 and 0.0446, respectively. Experimental results on the TIMIT and Harvard Sentences datasets, enriched with diverse noise sources from the DEMAND database, yielded high intelligibility scores with a short-time objective intelligibility (STOI) of 0.8345, word recognition score (WRS) of 99.2636, and an near correlation coefficient (LCC) of 0.9607, underscoring the model's capability to enhance speech perception in noisy CI environments, ensuring a balance between model size and speech quality, and surpassing the existing state-of-the-art techniques.

INDEX TERMS Cochlear implant, deep learning, sound coding strategy, speech enhancement, transformer.

I. INTRODUCTION

Automatic speech recognition (ASR), a cornerstone of modern technological advancements, plays a pivotal role in enriching human-machine interactions. As a fundamental component of our daily routines, automatic speech recognition (ASR) influences our methods of communication, information retrieval, and navigation through complex healthcare systems. The utility of ASR transcends simple interaction facilitation, extending into diverse areas including voice-activated services [1], speech-to-text conversion [2], user authentication [3], and notably, biomedical investigations [4], [5]. Moreover, the ASR and the speech processing techniques contribute to converting spoken language into either 1D (text) or 2D (spectrogram) data, enabling seamless integration across diverse digital platforms. Today, ASR is

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embedded in numerous devices such as smartphones, smart home devices, and interactive virtual assistants, all of which rely on voice commands to perform tasks, thereby enhancing user convenience by supporting hands-free operations.

ASR is also integral to authentication frameworks, enhancing the security of confidential data. Techniques like watermarking and steganalysis [6], [7], [8], [9], [10], and speech biometrics [11] leverage ASR to validate and secure speech communications. Beyond its communication applications, ASR serves key roles across various domains. For instance, *speaker recognition* has become indispensable in healthcare, facilitating quick and secure patient identification [12], crucial for accessing and managing medical records or executing health interventions. Meanwhile, *event recognition* through ASR transforms sectors from security to healthcare; in security, it enhances monitoring by detecting specific acoustic events, and in healthcare, it aids in the early detection and management of medical conditions by analyzing speech in challenging environments [13], assessing speech impairments like dysarthria [5], and identifying abnormal heart sounds for early preventative care [14]. Additionally, *source separation* plays a vital role in distinguishing individual audio sources within complex environments [15], valuable not only in entertainment but also in biomedical contexts where it helps interpret physiological sounds.

In biomedical field, individuals with severe or profound sensorineural hearing loss who retain healthy auditory nerve fibers may benefit from receiving a cochlear implant (CI) to partially restore their hearing. A CI consists of an electrode array implanted in the cochlea and a wearable sound processor that converts acoustic signals into electrical stimulation patterns delivered to the auditory nerve fibers. However, there is a significant performance gap between individuals with normal hearing and CI users for various auditory tasks. Electric stimulation has limitations in effectively conveying sound information, leading researchers to explore improved sound-coding strategies for CIs. Evaluating the potential benefits of new strategies often requires extensive testing with implanted volunteers; however, the high variability in performance among CI users makes it challenging to generalize the results. For example, deep auto encoder (DAE) was applied in [16], for noise reduction and speech enhancement using only clean speech. The study, a denoising process was introduced by training the DAE with noisy-clean pairs. Greedy layer-wise pretraining and fine-tuning were used, treating each layer as a one-hidden-layer in DAE with noisy-clean pairs. The trained DAE serves as a filter for noisy speech. Yet, [17] introduces WaveCRN, an efficient end-to-end speech enhancement model using a convolutional neural network (CNN) module for capturing speech locality and a stacked simple recurrent units (SRUs) module for sequential properties. Unlike traditional recurrent neural network (RNN) or long short-term memory (LSTM), SRU allows for parallel calculation with fewer parameters. A novel restricted feature masking enhances noise suppression within hidden layers. However, all the previously mentioned solution have not been tested for compatibility with small devices, such as CIs.

This paper introduces a novel noise suppression technique aimed at enhancing speech intelligibility for CI users by optimizing deep learning (DL)-based CI algorithms. Leveraging temporal convolutional networks (TCNs), multi-head Attention (MHA), and bidirectional gated recurrent unit (Bi-GRU), the approach efficiently balances model size and speech quality, particularly in noisy environments.

The contributions of this work are summarized as follows:

1) **Integration of MHA, TCN, and Bi-GRU for advanced sequence modeling:** This study introduces a novel hybrid architecture that combines MHA, TCN, and Bi-GRU to capture both local and global temporal dependencies in speech signals. This innovative design enables enhanced noise suppression and intelligibility improvements, a combination not previously explored in CI-related research.
2) **Custom modules inspired by CI signal processing:** Incorporates multiple strategies (MHA with TCN, and MHA-GRU with TCN) tailored to address the unique auditory requirements of CIs. These include an encoder inspired by the auditory signal processing pipeline of CIs and a separator module based on mask estimation principles. The encoder is designed to extract critical auditory features efficiently, while the separator module ensures precise signal reconstruction by dynamically estimating optimal masks for speech and noise separation. Each strategy balances key trade-offs, such as maintaining computational efficiency for real-time applications, preserving speech intelligibility, and minimizing distortion. By addressing these trade-offs, the framework provides a robust and practical solution for enhancing CI performance in noisy environments.
3) **Dual evaluation framework:** A comprehensive evaluation of the model is conducted across two domains: (i) the electrodogram domain, using linear correlation coefficient (LCC), and (ii) the speech domain, using short-time objective intelligibility (STOI) and word recognition score (WRS). This dual evaluation provides robust validation of the model's effectiveness in improving CI user experiences.
4) **Optimized for real-time inference:** The proposed model achieves a compact size of 788K parameters, balancing computational efficiency and high performance. This ensures feasibility for real-time applications, addressing a critical challenge in CI processing.
5) **Robust performance in realistic noise scenarios:** The model's robustness is validated using diverse real-world noise conditions from the DEMAND dataset combined with the TIMIT and Harvard sentences datasets. This comprehensive testing demonstrates the model's applicability to practical, noisy environments, setting a new standard for CI-related studies.

The proposed framework advances current speech enhancement techniques for CIs by innovatively integrating

TCN, MHA, and Bi-GRU layers. Unlike existing methods that predominantly rely on singular approaches such as CNNs or traditional RNNs, this method combines complementary architectures to capture both local and global temporal dependencies in speech signals. The compact design, with only 788K parameters, ensures real-time applicability without compromising performance. Experimental evaluations on diverse datasets, enriched with real-world noise conditions, underscore the robustness and generalizability of the approach, making it a significant leap forward in the field of CI speech enhancement. By benchmarking against state-of-the-art models, the proposed technique demonstrates marked improvements in intelligibility and noise suppression, setting a new standard for DL-based CI strategies.

The paper is structured as follows: Section II reviews related work on CI coding strategies and DL-based noise reduction methods. Section III introduces the background of existing CI stimulation strategies, focusing on traditional and DL-based coding techniques. Section IV details the proposed denoising framework, including architectural innovations such as TCN, MHA, and Bi-GRU. Section V discusses the experimental setup, datasets, evaluation metrics, and results. Finally, section VI provides a comprehensive conclusion and suggests directions for future research.

II. RELATED WORK

Convolutional neural networks (CNNs) have gained significant attention in speech-enhancement systems owing to their ability to exploit the temporal and spectral characteristics of speech signals. These networks have been effectively employed to reduce noise and improve speech quality, particularly in challenging low signal to noise ratio (SNR) scenarios. An important aspect of CNN-based speech enhancement is the incorporation of skip connections. Skip connections provide additional pathways for information flow, enabling the network to focus on relevant noise properties. For example, Zheng et al. [18] investigated the impact of skip connections in CNN architectures for denoising speech signals. The results indicate that skip connections improve the performance of CNNs, particularly in the presence of non-stationary noise. The correlation distance skip connection-denoising autoencoder (CDSK-DAE) introduces a novel approach that incorporates skip connections on both the encoder and decoder sides of the autoencoder [19]. This architecture allows for the transfer of speech information from the input to the model, thereby enhancing the dependency between the latent and enhanced features. The CDSK-DAE method also introduces a correlation distance measure in the objective function, aiming to align the features of the model with the desired target features. The experimental results demonstrate that CDSK-DAE outperforms conventional and state-of-the-art models in noisy environments, significantly reducing the word error rate (WER) in both seen and unseen noisy conditions. Moving on, Wang et al. in [20] explored the use of fully convolutional neural networks (FCN) to improve speech intelligibility in simulated electric and acoustic stimulation (EAS) devices. The study compared an FCN with traditional speech enhancement methods and evaluated its effectiveness in noisy environments. Objective evaluations and listening tests showed that the FCN outperforms other methods in enhancing speech intelligibility for both normal and vocoded speech. This research confirms that FCN is a promising approach for EAS devices, offering potential benefits to users under noisy conditions. This study highlights the importance of speech enhancement in EAS technology and suggests the integration of FCN into EAS processors to enhance user experience.

Denoising autoencoders, another popular DL model, have gained significant attention in the field of speech processing enhancement and denoising because of their ability to learn compact representations and effectively remove unwanted noise. The denoising variational autoencoder (VAE) proposed in [21] utilizes a deep generative model combined with unsupervised noise models to enhance speech robustly. By employing an encoder network, the denoising VAE estimates the latent vectors of clean speech from a mixture signal, enabling the model to capture the underlying structure of speech while removing noise. The use of a variational framework allows for the modelling of uncertainty, resulting in a more accurate speech reconstruction and improved robustness under various conditions. In addition, the multi-layered convolutional neural network (MLCNN)-based auto encoder-decoder leverages the power of DL and specifically designed convolutional layers to extract meaningful features from noisy audio signals [22]. By utilizing mel-frequency cepstral coefficients (MFCC) as input, the MLCNN model achieves high accuracy in enhancing audio signal quality, outperforming other methods in terms of performance evaluation metrics. The proposed denoising autoencoder with multi-branched encoders (DAEME) model in [23] introduces a novel method for speech enhancement, specifically targeting challenges related to mismatched noisy conditions and focused noise types. It utilizes a decision-tree (DT)-based multi-branched encoder, enabling precise mapping from noisy to clean speech. Experimental results show that DAEME outperforms baseline models in objective metrics, ASR, and subjective listening tests. The DT-based encoder enhances the model's adaptability to both seen and unseen noise types, leveraging DT knowledge for optimal performance across various noisy environments.

In biomedical applications, DL has emerged as a powerful technology for developing coding strategies in CIs to enhance speech intelligibility in noisy environments. Numerous studies have proposed innovative approaches that leverage deep neural networks (DNNs) to significantly improve the performance of CI sound processing. For example, Lai et al. [24] have investigated the use of a deep denoising autoencoder (DDAE) approach to improve the intelligibility of vocoded speech in CI simulation. This study aimed to enhance the speech perception of CI users under noisy conditions. The DDAE-based noise reduction (NR) method was compared to conventional single-microphone NR approaches using noise vocoder simulation. Objective evaluations and listening tests show that the DDAE-based NR approach achieves higher intelligibility scores than traditional methods, especially under non-stationary noise distortion conditions. These findings suggest that incorporating DDAE-based NR into CI processors could offer significant benefits to CI users in noisy environments. This research contributes to ongoing efforts to develop effective NR strategies for enhancing speech perception in CI users. Moving forward, in [25], a DNN was trained to estimate CI electric stimulation patterns from raw audio data, effectively replacing traditional CI sound coding strategies. The study demonstrates that the proposed approach improves speech intelligibility under noisy conditions without compromising latency. It achieves this by capturing the complex relationships between the input audio features and desired stimulation patterns, facilitating speech coding and denoising. Gajecki et al. [26] proposed replacing traditional CI sound-coding strategies with a DNN-based method, called DeepACE. The network accurately codes acoustic signals while automatically removing interfering noise, resulting in higher speech intelligibility scores compared with baseline algorithms. The proposed strategy shows potential for enhancing speech understanding under challenging acoustic conditions. Later, the same researcher in [27], suggested a fused deep denoising sound coding strategy is introduced for bilateral CIs. By connecting two monaural deep denoising algorithms through fusion layers, this approach effectively reduces noise and improves the learning generalization. The experimental results demonstrate higher interaural coherence, superior noise reduction, and enhanced speech intelligibility compared to baseline methods. This strategy holds promise in enhancing the listening experience of bilateral CI users. Mamun et al. in [28], present a CNN-based speech enhancement algorithm designed specifically for CI recipients. The algorithm leverages CI auditory features and addresses real-time application delays. The experimental results show significant improvements in speech intelligibility compared with existing baseline systems. The proposed approach has the potential to enhance the speech perception of CI users in natural environments. Furthermore, Huang et al. [29] introduced ElectrodeNet, a DL-based sound-coding strategy for CIs. The strategy replaces conventional envelope detection with DNNs, CNNs, or LSTM networks trained using clean speech processed by the advanced combination encoder (ACE) strategy. Objective and subjective evaluations demonstrated strong correlations between ElectrodeNet and ACE in terms of understanding speech. The proposed ElectrodeNet-CS strategy combines envelope detection and channel selection (CS) in a single network, achieving comparable or better speech intelligibility than ACE.

Recent advances in the use of Transformer-based models within DL have significantly propelled the field of ASR [30], subsequently enhancing CI strategies. The Conformer architecture, which combines the strengths of Transformers and CNNs, has emerged as a state-of-the-art approach, achieving superior performance in ASR. Other notable Transformer models include speech Transformer, which replaces traditional RNN and LSTM approaches with ***Attention*** mechanisms [31], [32], resulting in lower training time and memory usage. Figure 1 illustrates a comparison of audio and speech models based on training mode, namely supervised, self-supervised, and unsupervised.

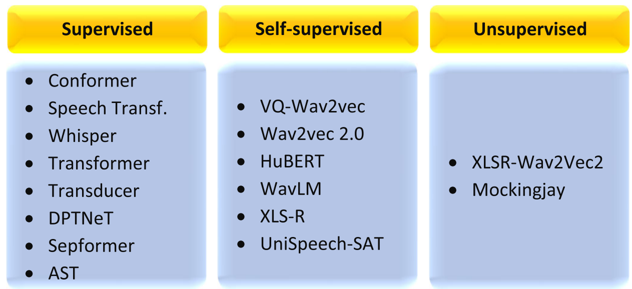


FIGURE 1. Contrasting audio and speech models by their respective supervision methodologies [33].

In the context of speech classification, Transformer-based models have demonstrated their capabilities. audio spectrogram Transformer (AST) offers a convolution-free approach, capturing long-range context using a Transformer encoder and achieving excellent performance in audio/speech classification tasks. Mockingjay, an unsupervised speech representation model, employs multiple layers of bidirectional Transformer pretrained encoders, enabling effective speech classification [33]. Saleem et al. [34] presents a time domain speech enhancement model using a combination of CNN and a time-attention Transformer (TAT). The proposed model aims to improve the quality and intelligibility of noisy speech by processing the waveform directly without explicit feature extraction or domain transformation. The CNN-based encoder-decoder framework incorporates 1D-time domain dilated residual blocks to preserve detailed features, and the TAT model integrates a time-attention mechanism to capture long-term dependencies in the speech signal. Experimental results demonstrate that the proposed model outperforms recent deep neural networks, achieving significant improvements in speech intelligibility and quality. The model shows promise for real-time applications such as hearing aids and ASR.

Table 1 summarizes the best performance achieved by related DL-based CI studies across specific datasets, along with comparative methods and relevant comments.

III. BACKGROUND

This section provides a brief introduction to the most commonly used CI stimulation strategies, including DeepACE, the latest strategy for CI stimulation, which serves as our benchmark method.

TABLE 1. A summary of related DL-based CI study performances.

Ref.	DLM	Dataset	Best performance	Compared to	IMP (%)	Comment
[20]	FCN	TMHINT	STOI (engine noise) 0.64 at -3 dB STOI (street noise) 0.71 at -3 dB	MMSE, DDAE	STOI (engine noise) +0.05 at -3 dB STOI (street noise) +0.05 at -3 dB	DDAE method is better than minimum mean square error (MMSE).
[28]	CNN	UT-Drive, TIMIT	ECM= 99.20%	Log-MMSE, Wiener-as, Wiener-wt	ECM= +0.8, ECM= +0.7, ECM= +2.2	These results pertain to non-causal speech enhancement in the presence of Nissan Sentra noise.
[29]	DNN, CNN, LSTM	TIMIT, TMHINT	MSE-based-STOI $\in [0.0004 - 0.0007]$ mean square error (MSE)-based-NCM $\in [0.0041 - 0.0066]$ LCC-based-STOI $\in [0.9937 - 0.9949]$ LCC-based-NCM $\in [0.9898 - 0.9919]$	ACE	STOI $\in [0.0013 - 0.0021]$ STOI ≤ 0.0015 in quiet conditions NCM ≤ 0.0024	Electrodenet model was compared to the original ACE strategy using a vocoder to calculate the STOI and NCM scores. The comparison of these scores was evaluated using the MSE and LCC metrics.
[25]	AE	-FCEC[1] -HSM -ICRA7 -CCITT noise	STOI = 0.78	ACE	STOI = -0.02	The approach allows reducing the processing complexity of the ACE while performing noise reduction for CIs.
[26]	AE	- LibriVox - TIMIT - HSM - DEMAND - Synthetic - SSN - ICRA7	STOI = 0.803	ACE	STOI = -0.004	The comparison conducted here is under quiet conditions. When noise is added, there is an improvement compared to ACE, and the STOI score is significantly high.
Our	TCN, MHA, Bi-GRU	TIMIT, Harvard sentences, DEMAND, TMHINT	STOI = 0.8145	Electrodenet_CS, FCN, MMSE, DDEA	STOI = +0.021 (0dB), STOI = +0.007 (1dB), STOI = +0.015 (5dB), STOI = +0.012 (10dB)	The proposed model demonstrates superior performance in improving STOI across varying noise levels, highlighting its robustness and adaptability in noisy environments. Unlike prior methods, it leverages a combination of TCN, MHA, and Bi-GRU, enabling effective denoising and enhanced intelligibility for CI signal processing.

Abbreviations: DL method (DLM), Improvement (IMP), accuracy (Acc).

A. EXISTING CI CODING STRATEGIES

The CI is a highly successful machine-brain interface that restores auditory perception for individuals with severe hearing impairment. While CI recipients can achieve good speech understanding in quiet environments, their performance is significantly worse than that of individuals with normal hearing in realistic listening situations, especially in the presence of noise. In the normal auditory system, sound is captured by the outer ear, transmitted through the middle ear, and eventually reaches the cochlea, a spiral structure in the inner ear. The cochlea contains hair cells that respond to sound waves and generate electrical signals that are transmitted through the auditory nerve fibers. These signals carry temporal and tonotopic information, enabling the identification and interpretation of sounds at higher neural levels. The primary cause of deafness is damage or loss of hair cells, which can result from various factors such as infections, trauma, exposure to loud noise, certain drugs, or physiological disorders. When hair cells are absent or extensively damaged, the transduction of sound-induced motion to neural action potentials is disrupted, leading to severe hearing loss.

CI bypass the damaged hair cells and directly stimulate the auditory nerve or residual neurons using electrical signals. The implant consists of an external part, including a behind-the-ear device and an internal transmission coil, and an internal part, which contains electronics and an electrode array inserted into the cochlea [35]. The specifications of stimulation patterns are coded in a trans-cutaneous radio frequency (RF) transmission [36]. The implant decodes the stimulation specifications and delivers the electrical pulses through selected electrodes, activating neural structures near the electrode-neuron interface. The fitting process involves adjusting stimulation levels and parameters to optimize individual patient perception. Historically, CI objectives focused on improving speech intelligibility, which is influenced by spectral and temporal characteristics of the acoustic signal. Spectral information is coarsely coded through multi-channel representation, whereas temporal information is classified into the speech envelope, periodicity, and temporal fine structure (TFS). TFS is responsible for perceiving pitch, localizing sounds, and segregating sound sources. However, pitch perception with CIs is poor due to limitations in spectral resolution, imprecise coding of temporal cues, and the inability to accurately perceive TFS. Current research aims to enhance pitch perception, especially for music and tonal languages [37].

Different stimulation strategies convert sound signals into electrical stimuli or Electrodograms that activate the auditory nerve, as illustrated in Figure 2. Initially, these strategies

[1] First clarity enhancement challenge.

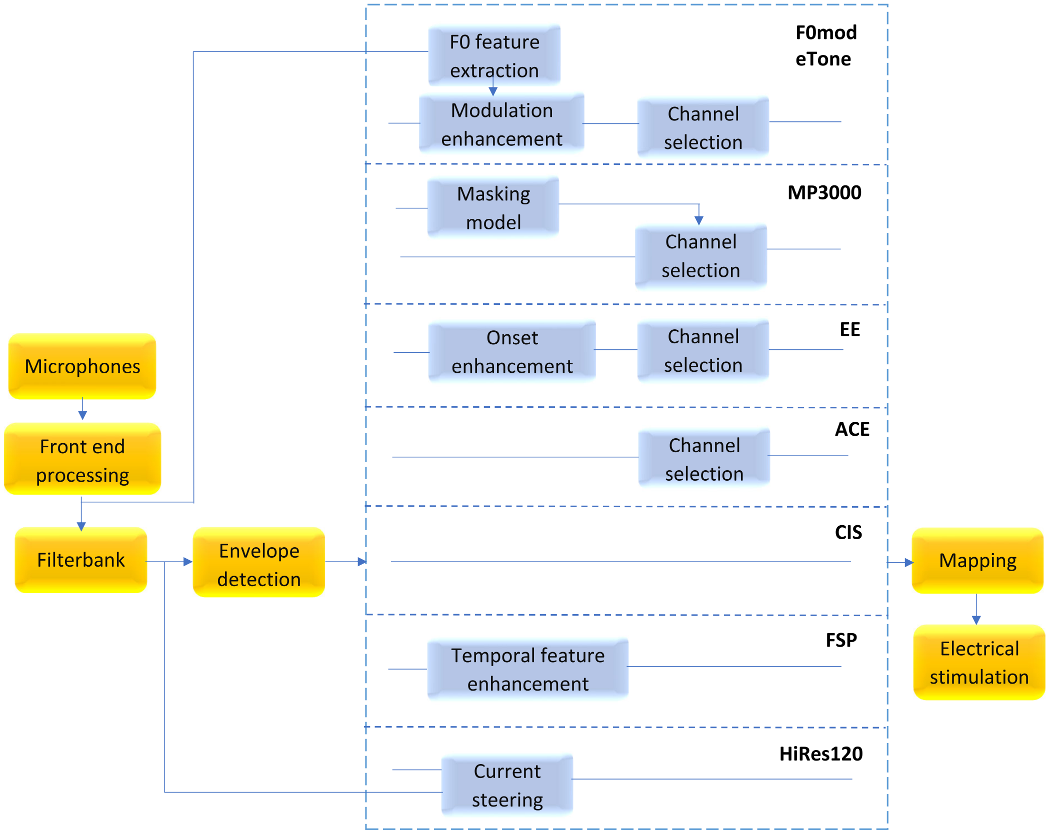


FIGURE 2. Block diagram of the most used CI stimulation strategies. Shared components are displayed on the left and right sides, whereas the elements specific to each strategy are enclosed within the dashed box in the center [37].

focused on feature extraction, estimating crucial speech signal characteristics such as the fundamental frequency (F_0) and formants (F_1 and F_2). However, these early methods result in poorer speech understanding compared to newer schemes and are rarely used in current commercial processors. One widely used strategy is continuous interleaved sampling (CIS). CIS involves a bank of band-pass filters or a fast Fourier transform (FFT) to analyze the input sound signal. The filter bank assigns filters to intracochlear electrodes based on tonotopic organization. The magnitude of the envelope in each channel is determined, and the stimulation levels are adjusted based on user-specific functions. The transformed magnitudes modulate carrier waves of electrical pulses, and a fixed stimulation carrier rate is used. CIS faithfully represents the temporal speech envelope, enabling effective transmission of envelope information for speech perception. Besides, other four most commonly used strategies are ACE, MP3000, fine structure processing (FSP), and high resolution (HiRes120). These strategies focus on better spectral representation, distribution of stimulation, and temporal representation of the input signal. ACE and MP3000 employ channel selection based on spectral features, while FSP enhances temporal features. HiRes120 combines temporal feature enhancement with current steering to improve spatial precision [37]. The strategies mentioned are utilized in commercial processors by leading manufacturers, including Cochlear, Advanced Bionics, Med-El, and Oticon Medical/Neurelec.

B. ADVANCED COMBINATION ENCODER (ACE) STRATEGY

The ACE strategy is an enhanced version of the CIS strategy that aims to reduce channel interaction and minimize the risk of simultaneous activation of all electrodes. It follows the "N of M" principle, stimulating a smaller number of spectrally dense channels N per cycle compared to the total available channels M. Subject tests have shown a preference for ACE over CIS, with a 30% higher preference rate [38].

As described in Figure 3, the CI microphone captures the acoustic signal and samples it at a rate of 16 kHz. Subsequently, a filter bank is employed using a 128-point FFT with a typical hop size of 32 points. This process introduces an algorithmic latency of 2ms, which may vary based on the channel stimulation rate (CSR). Each spectral band E_k $(k = 1, \ldots, M)$ is then assigned to a single electrode, representing a distinct channel. An estimation of the desired envelope is calculated for each spectral band. From the M channels in each audio frame, only the N channels with the highest spectral energy out of the total M channels in each stimulation cycle are chosen for stimulation. Typically, M and

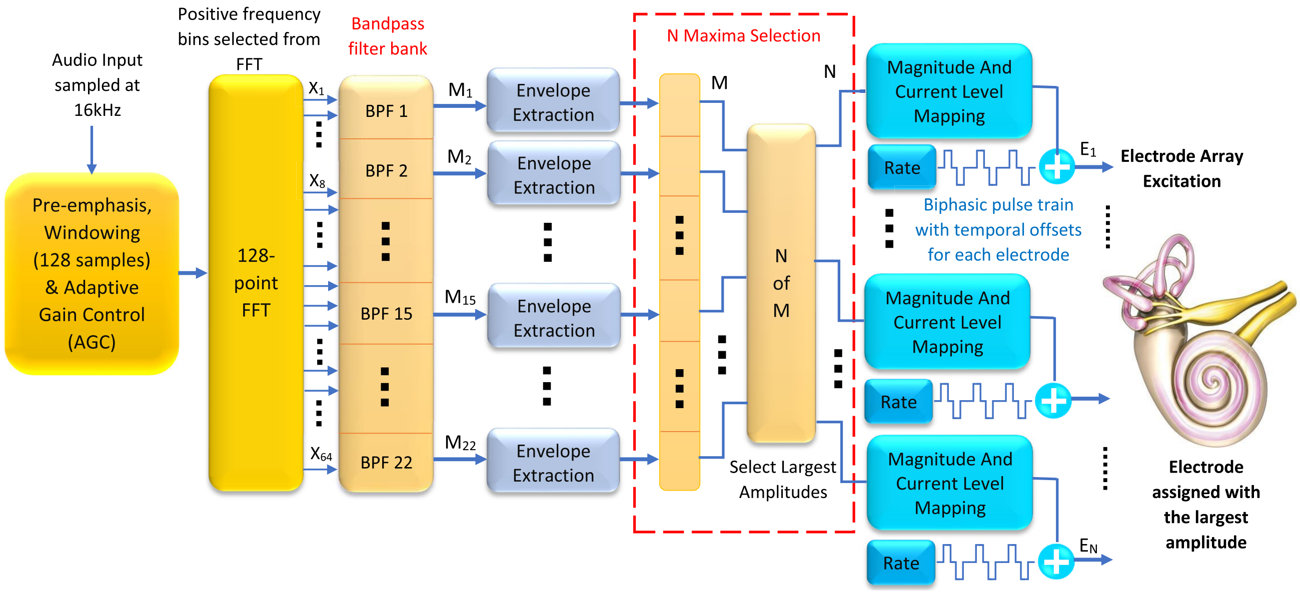


FIGURE 3. Block diagram of the ACE strategy as presented in [38].

N have values of 22 and 8, respectively. The selected bands undergo non-linear compression using a loudness growth function (LGF) defined by Equation 1.

$$p_k = \frac{\ln(1 + \rho \cdot \frac{E_k - s}{m - s})}{\ln(1 + \rho)} \tag{1}$$

Here, p_k represents the compressed value, E_k denotes the spectral band, and s, m, and ρ are calculated from clinical and sound processor parameters. If E_k falls below the base level s, p_k is set to 0, while if it exceeds the saturation level m, p_k is set to 1. Finally, in the last stage of the sound coding strategy, p_k is mapped to the subject's dynamic range between threshold levels (THLs) and most comfortable levels (MCLs) for electrical stimulation. Sequential stimulation occurs for the N selected electrodes in each audio frame, constituting one stimulation cycle. The CSR determines the number of cycles per second. Finally, the representation of the current applied to each electrode over time is known as an electrodogram.

C. FROM ADVANCED COMBINATION ENCODER (ACE) TO DEEPACE

The evolution from the traditional ACE strategy, an enhanced version of the CIS strategy, to DeepACE represents a significant leap forward in signal processing techniques for CIs. ACE, aimed to minimize channel interaction and reduce the risk of simultaneous electrode activation. It followed the principle of stimulating a smaller number of spectral dense channels per cycle compared to the total available channels, leading to improved user preference rates over CIS. ACE involved capturing acoustic signals, applying a filter bank with an FFT, selecting channels based on spectral energy levels, and compressing selected bands using a loudness growth function before stimulation.

DeepACE as presented in [26], on the other hand, signifies a transformation in signal processing methodologies by integrating DL into the process of generating stimulation signals for CIs. Designed to estimate the output of the loudness growth function using raw audio input and predicting denoised electrodograms, DeepACE operates independently of individual CI fitting parameters while maintaining the 2ms total algorithmic delay characteristic of the standard ACE strategy. The architecture of DeepACE introduces novel features, such as the deep envelope detector (DED) which consists of a stack of 3 convolutional layers and PReLUs, and enhanced hyperparameter configurations, which enhance the model's ability to process and interpret audio signals more effectively. Figure 4 shows the block diagrams of both ACE strategy and its enhanced version, DeepACE, which integrates DL.

This evolution from ACE to DeepACE showcases the transformative power of DL in advancing signal processing capabilities for CIs, offering more efficient, accurate, and adaptable methods for generating stimulation signals and improving the overall user experience in auditory prosthetics.

IV. PROPOSED FRAMEWORK

Our proposed framework follows a systematic approach to enhance the prediction of electrodograms using DL for CIs. Firstly, the original vocoded clean ACE-electrodograms, serving as reference signals are reconstructed. Then, the resynthesized audio waveforms from different strategies and the reference signals are used to calculate the STOI measure. This measure provides an objective assessment of the intelligibility of the speech signals and is commonly used to evaluate speech enhancement algorithms. The steps illustrated in Figure 5 are outlined as follows:

- **Model training:** With the training and validation datasets prepared, the proposed DL model is trained to predict electrodograms from noisy input audio. The model architecture is tailored for sequence processing tasks. During training, performance is monitored using validation metrics, and the model achieving the best performance based on the training/validation loss is saved. This model is considered the optimal candidate for testing, ensuring robustness in unseen noisy conditions.

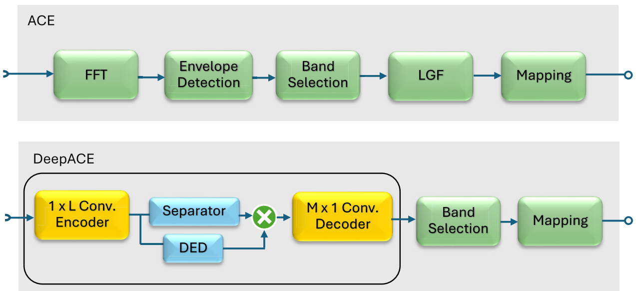


FIGURE 4. Evolution from ACE to DeepACE strategy. The training regimen for DeepACE involves batches of audio segments, each lasting 4 seconds, and is trained for a maximum of 100 epochs. Hyperparameters are optimized through empirical testing to improve model performance compared to previous versions. With the inclusion of the DED and the *separator*, DeepACE streamlines the dimensionality reduction process between the encoder and decoder modules, aligning the number of bands for stimulation and extracting crucial features from the encoded signal.

- **Model testing:** After the training phase, the saved model is evaluated on a separate test set comprising noisy.wav audio signals mixed with various types of environmental noise, unseen during training. The model predicts the corresponding electrodograms from these noisy audio inputs.
- **Electrodogram domain evaluation:** To assess the accuracy of the predicted electrodograms, an evaluation is first performed in the electrodogram domain. The predicted electrodograms are compared to the clean electrodograms generated by the traditional ACE strategy for the original clean audio. This evaluation quantifies the similarity between the predicted and ground truth electrodograms, typically using the LCC metric to measure reconstruction quality.
- **Speech domain evaluation:** The second stage of evaluation takes place in the speech domain. Using a vocoder, the predicted electrodograms are converted back into audio waveforms to synthesize a raw audio signal. This synthesized audio is then assessed for intelligibility and perceptual quality. To quantify intelligibility, the STOI and WRS metrics are used, providing an objective measure of how well the synthesized speech matches the clean audio in terms of clarity and intelligibility.

This dual evaluation, in both the electrodogram and speech domains, offers comprehensive insights into the model's effectiveness in real-world noisy environments.

A. BASELINE MODEL

Our baseline method is based on the deep denoising sound coding strategy described in [26]. This approach, named DeepACE, is designed to estimate the denoised electrodograms by taking raw audio input. It is independent of individual CI fitting parameters and maintains the standard ACE strategy's $2ms$ total algorithmic delay. The baseline model of DeepACE, as depicted in Figure 4, introduces two primary architectural innovations: **(i)** Firstly, it includes a DED positioned in the skipping path of the original DeepACE model [25]. The DED performs dimensionality reduction between the encoder and decoder modules, matching the number of bands to be stimulated and extracting essential features from the encoded signal. The DED module consists of three consecutive 1-D convolution layers stacked together. **(ii)** Secondly, the separator block is responsible for separating the target speech signal from the background noise or interfering signals to enhance speech intelligibility for CI users, the estimated mask must have the same dimensions as the LGF output.

B. PROPOSED STRATEGIES

The proposed model aims to enhance the CI coding strategy block through the implementation of three key components:

- **TCN block:** In the landscape of DL, the exploration of TCNs as a formidable alternative to conventional recurrent architectures has sparked a paradigm shift. The underpinning concept of TCNs lies in their utilization of 1D temporal convolutions to unravel intricate patterns and dependencies within sequential data. This mathematical framework, exemplified by the convolutional operation [39]:

$$y[t] = f\left(\sum_{i=1}^{k} w[i] \cdot x[t-i] + b\right) \tag{2}$$

where $y[t]$ denotes the output at time step t, f signifies the activation function, w represents filter weights, $x[t-i]$ embodies inputs at prior time steps, b stands for the bias term, and k represents the kernel size, forms the backbone of TCNs. This mechanism equips TCNs with the ability to efficiently capture temporal nuances and intricate dependencies within sequences, thereby offering a fresh perspective on memory retention and information propagation in sequence modeling tasks. The allure of TCNs extends further to their exceptional capacity in retaining extended historical information, surpassing the capabilities of traditional recurrent architectures. This augmented memory retention is instrumental for tasks mandating a profound understanding of historical contexts over prolonged temporal spans, underscoring the superiority of TCNs in handling complex sequence modeling tasks with finesse and efficacy.

The separator module is inspired by the Conv-TasNet system [40], is designed to calculate a multiplicative function, commonly referred to as a mask, for each

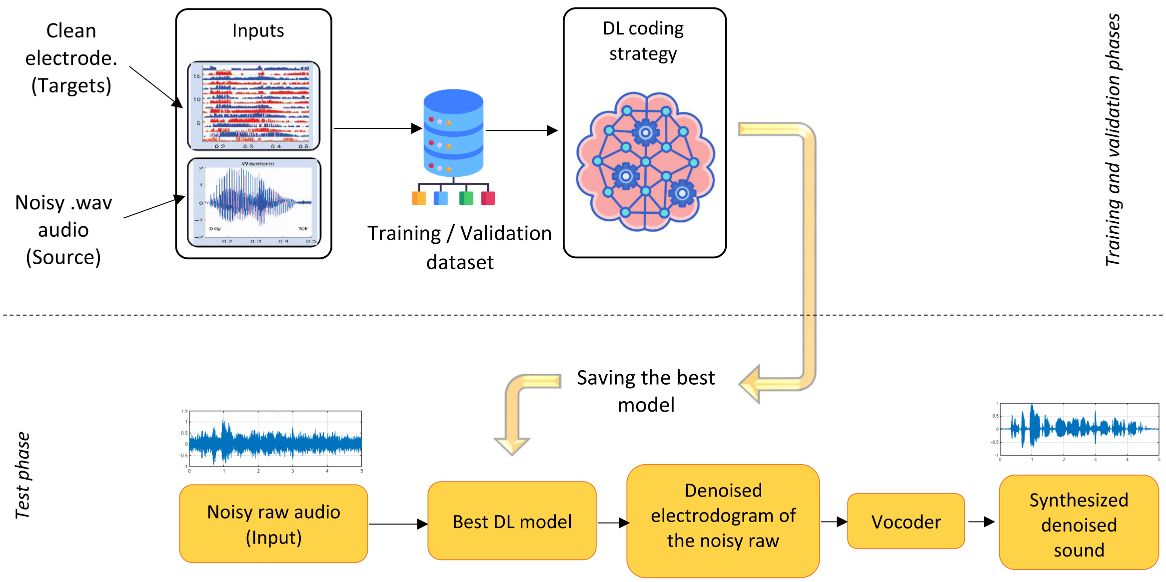


FIGURE 5. **DL framework for electrodogram prediction and evaluation in noisy environments.**

target source in the input signal. The separator leverages a TCN to estimate these masks effectively. The TCN separation module processes the encoded features from the encoder output and produces masks that isolate and enhance the target sources. By applying these masks, the system is able to selectively extract and reconstruct individual components from complex auditory signals, ensuring accurate and efficient separation.

- **Multi-head attention (MHA) layer:** The Attention approach allows the model to excel in capturing both short-term and long-term dependencies, context, and relationships within speech data, making it particularly effective for ASR tasks, especially in challenging environments. Figure 6 illustrates the structure and key components of the Attention layer. In the context of CIs, Attention Transformers can significantly enhance speech recognition and processing by learning to detect and amplify subtle speech patterns that traditional ASR methods might overlook, particularly in noisy environments or with distorted inputs. The Attention mechanism can be mathematically represented by the Equation [30]:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right).V \tag{3}$$

Here, Q, K, and V stand for queries, keys, and values, respectively. In the context of ASR for CIs, queries could represent the expected features of clean speech, while keys are derived from the actual received speech signals. These keys are used to compute Attention weights, indicating the importance of different speech segments. Values represent the processed speech data that, after applying Attention weights, are aggregated into the final recognized output. Additionally, d_k is the dimensionality of the keys. This mechanism allows the model to focus on the most relevant parts of the speech signal, thereby enhancing the accuracy and intelligibility of the recognized speech, which is crucial for the effectiveness of CIs.

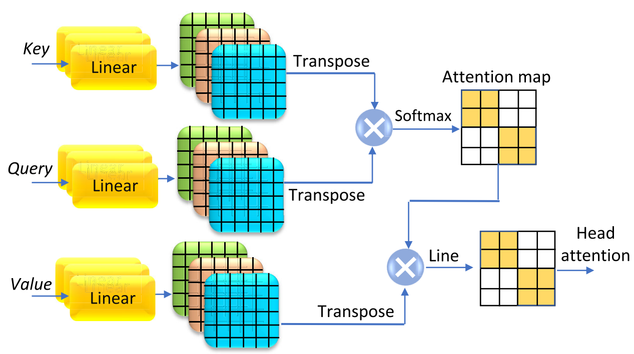


FIGURE 6. **Visual structure of attention mechanism.**

MHA, a fundamental concept in modern neural network architectures, revolutionizes conventional attention mechanisms by enriching the model's ability to process information across multiple perspectives simultaneously. By introducing MHA, each equipped with distinct learned linear projections transforming queries, keys, and values into diverse dimensions, the model gains the capacity to explore intricate relationships within the data in parallel. The core equation governing MHA is [41]:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(head_1, \ldots, head_h)W^O \tag{4}$$

where each $head_i$ is computed as:

$$head_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{5}$$

Here, the projections are defined by parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, and $W^O \in \mathbb{R}^{h \times d_v \times d_{\text{model}}}$.

In this innovative framework, rather than relying on a singular attention function with fixed-dimensional inputs, the queries, keys, and values undergo individual linear transformations h times. These transformed representations are then processed concurrently through attention mechanisms, generating output values that encapsulate a broader range of information. Subsequently, these outputs are fused together through concatenation and further projection, culminating in a comprehensive and enriched representation of the original data. This strategic design choice ensures that the model can efficiently harness the benefits of MHA without exponentially increasing computational complexity, thus paving the way for more sophisticated and effective neural network architectures.

- **Gated recurrent units (GRU) layer:** It is a type of RNN architecture designed to address the vanishing gradient problem in traditional RNNs. gated recurrent unitss (GRUs) are similar to LSTM units but have a more straightforward architecture with fewer parameters, making them computationally less expensive. Figure 7 illustrates the detailed architecture of a GRU unit.

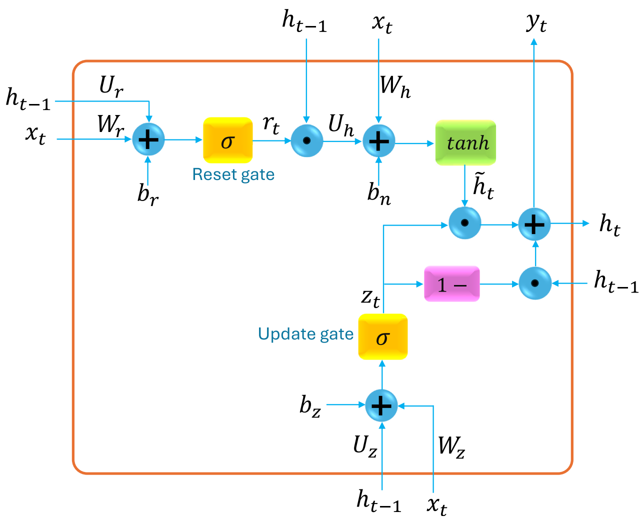


FIGURE 7. In-depth description of a GRU cell's architecture [42]. GRUs contain two gates, the reset gate and the update gate. These gates control the flow of information within the GRU cell. The reset gate determines how much of the past information to forget, while the update gate regulates how much of the new information is integrated into the cell state. This gating mechanism enables GRUs to capture long-range dependencies in sequential data effectively.

In a GRU, the update gate z, reset gate r, candidate hidden state $\tilde{h}_t$, and new hidden state h_t are computed as follows [43]:

The update gate is calculated as:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \tag{6}$$

The reset gate is computed as $r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$, the candidate hidden state is determined by $\tilde{h}_t = \tanh(W \cdot [r_t \odot h_{t-1}, x_t] + b)$, and the new hidden state is updated according to $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$.

The Bi-GRU extends the idea of the GRU by processing input sequences in both forward and backward directions simultaneously as illustrated in Figure 8. By combining two GRU networks—one processing the input sequence from the beginning to end and the other processing it in reverse—Bi-GRU captures dependencies from both past and future contexts. The forward hidden state $\overrightarrow{h}_t$, backward hidden state $\overleftarrow{h}_t$, and final Bi-GRU hidden state h_t are determined as $\overrightarrow{h}_t = \text{GRU}(x_t, \overrightarrow{h}_{t-1})$ and $\overleftarrow{h}_t = \text{GRU}(x_t, \overleftarrow{h}_{t+1})$ [44], respectively.

These mathematical descriptions encapsulate the core computations of GRU and Bi-GRU models, allowing them to efficiently capture dependencies in sequential data by updating hidden states using input and contextual information from both preceding and succeeding time steps. Bi-GRU is particularly useful in tasks where contextual information from both directions is essential, such as natural language processing (NLP) tasks like machine translation and sentiment analysis. By leveraging information from both directions, Bi-GRU models can better understand and capture complex patterns in sequential data, leading to improved performance on a wide range of sequential learning tasks.

1) FIRST PREDICTION STRATEGY: MHA WITH TCN

The first scenario introduces a groundbreaking neural network architecture that merges the capabilities of TCN with MHA in a parallel framework, with the fusion of their outputs through concatenation serving as a pivotal element. TCNs specialize in capturing local dependencies within sequential data, while MHA mechanisms excel at deciphering intricate relationships across different segments of the input sequence simultaneously. By operating TCN and MHA in parallel and merging their outputs, the model achieves a harmonious blend of local and global context representations, leading to a holistic understanding of temporal data.

This integration not only empowers the model to harness the strengths of both architectures for enhanced temporal modeling but also enables sophisticated pattern recognition within sequential data, propelling it towards more accurate predictions and analyses. The combined architecture efficiently handles long-range dependencies, ensuring the retention of crucial contextual information over extended sequences, while offering a versatile and adaptable framework suitable for diverse applications requiring nuanced temporal modeling, such as natural language processing NLP and time series analysis. Through the parallel execution and concatenation of TCN and MHA outputs, the model enables intricate pattern identification and fosters in-depth analysis of sequential data across our tasks.

The model's mathematical formulation begins by passing the input feature representation $\mathbf{X}$, produced by a 1D

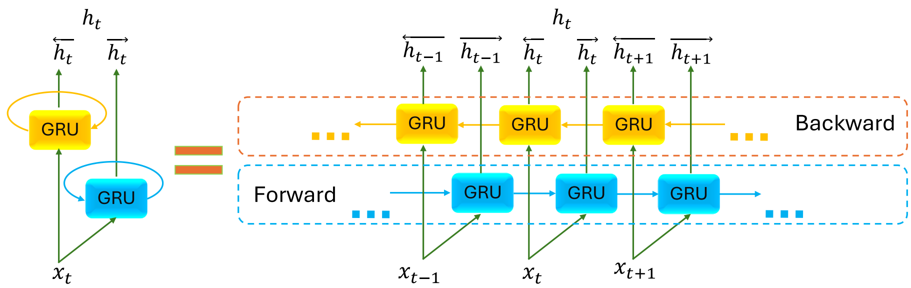
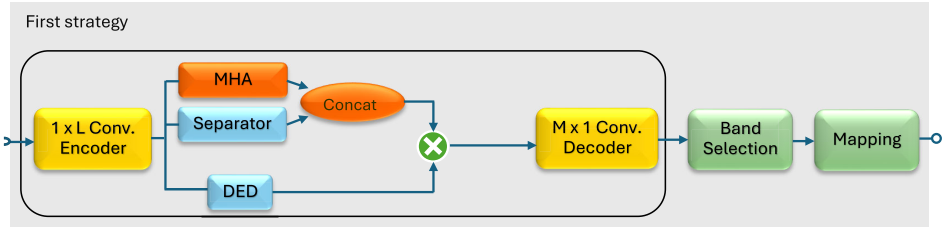
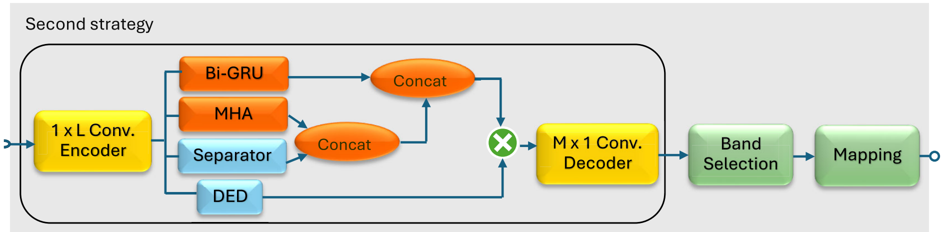


FIGURE 8. Configuration of a bidirectional GRU architecture [42].



(a)



(b)

FIGURE 9. Overview of developed DL strategies for CI speech denoising.

convolutional encoder, into three components: the DED, the separator, and the MHA mechanism.

The deep envelope detector applies an operation on $\mathbf{X}$ to extract envelope-based features, resulting in an output $\mathbf{D}$:

$$\mathbf{D} = \mathrm{DED}(\mathbf{X})$$

The separator and MHA modules, operating independently on $\mathbf{X}$, produce outputs $\mathbf{S}$ and $\mathbf{M}$, respectively:

$$\mathbf{S} = \mathrm{Separator}(\mathbf{X}), \qquad \mathbf{M} = \mathrm{MHA}(\mathbf{X})$$

These outputs $\mathbf{S}$ and $\mathbf{M}$ are concatenated to form a combined representation $\mathbf{C}$:

$$\mathbf{C} = \mathrm{Concat}(\mathbf{S}, \mathbf{M}) \tag{7}$$

The final output $\mathbf{Y}$ is obtained by performing an element-wise multiplication between $\mathbf{C}$ and $\mathbf{D}$, integrating the envelope features with the concatenated output:

$$\mathbf{Y} = (\mathrm{Concat}(\mathrm{Separator}(\mathbf{X}), \mathrm{MHA}(\mathbf{X}))) \odot \mathrm{DED}(\mathbf{X})$$

Where $\odot$ denotes an element-wise multiplication operation. This formulation reflects the architecture's ability to combine TCN and MHA outputs in a way that supports sophisticated temporal modeling, enhancing its capability for accurate predictions and deep sequential data analysis, as illustrated in Figure 9 (a) with the integration of the MHA mechanism.

2) SECOND PREDICTION STRATEGY: MHA-GRU WITH TCN

The model in the second scenario ingeniously combines a diverse set of advanced components to comprehensively capture both local and global temporal dependencies within sequential data. By integrating an encoder for feature extraction, a TCN layer for long-range dependency learning, Bi-GRU layer for sequential understanding, and a MHA mechanism for selective focus, the model effectively merges spatial-temporal features and sequential information, enhancing predictive accuracy. Through parallel and concatenated operations, the model harmonizes local and global context representations, enabling a profound understanding of temporal data and intricate pattern recognition within sequential information.

In mathematical terms, the second strategy differs from the first after Equation 7, where the Bi-GRU is introduced to process $\mathbf{X}$ and produces an output $\mathbf{G}$:

$$\mathbf{G} = \mathrm{BiGRU}(\mathbf{X})$$

Next, the concatenated representation $\mathbf{C}_1$ is further concatenated with the output of the Bi-GRU, forming $\mathbf{C}_2$:

$$\mathbf{C}_1 = \mathrm{Concat}(\mathbf{C}, \mathbf{G})$$

The final output $\mathbf{Y}$ is obtained by performing an element-wise multiplication between $\mathbf{C}_1$ and $\mathbf{D}$, effectively combining the envelope features from the DED with the concatenated representations from the separator, multi-head attention, and Bi-GRU:

$$\mathbf{Y} = \mathbf{C}_1 \odot \mathbf{D}$$

Expressed as a single equation, the final output can be written as:

$$\mathbf{Y} = (\mathrm{Concat}(\mathrm{Concat}(\mathrm{Separator}(\mathbf{X}), \mathrm{MHA}(\mathbf{X})), \mathrm{BiGRU}(\mathbf{X})))$$
$$\odot\, \mathrm{DED}(\mathbf{X})$$

This formulation underscores the architecture's ability to integrate diverse components in parallel, capturing complex temporal dependencies and enabling nuanced analysis of sequential data, as shown in Figure 9 (b) with the integration of the Bi-GRU mechanism.

The proposed framework integrates TCNs, MHA, and Bi-GRU to address the challenges of speech intelligibility in noisy environments. TCNs efficiently model local temporal patterns through hierarchical 1D convolutions, ensuring precise noise suppression at fine time scales. MHA layers capture global dependencies by selectively focusing on critical speech components while filtering out irrelevant noise. Bi-GRU layers process sequences bidirectionally, capturing contextual information from both past and future frames, thereby enhancing the continuity and intelligibility of speech signals. Unlike conventional models such as CNN, which are limited in capturing long-term dependencies, or RNN which are computationally intensive for real-time applications, this hybrid architecture combines the complementary strengths of TCNs, MHA, and Bi-GRU to overcome these limitations. Its compact design ensures computational efficiency, making it ideal for real-time deployment in cochlear implant systems. The design is intended to achieve superior STOI and WRS performance, making it well-suited for real-world noisy environments.

V. EXPERIMENTAL RESULTS

This section evaluates the baseline and proposed scenarios to determine the optimal configuration for electrodigram denoising in the context of CI. It is important to note that additional scenarios, were also explored.

A. DATASETS AND METRICS

1) DATASETS

Three diverse datasets are used to evaluate the proposed strategies, enabling a comprehensive comparison with state-of-the-art methods and providing a robust assessment of the method's generalizability. The details of these datasets are provided below:

- **TIMIT:** The TIMIT database, widely used in speech processing research, contains recordings from 630 speakers, covering eight major American English dialects. Each speaker reads 10 phonetically rich sentences, resulting in 6,300 utterances. The dataset is organized into a training set of 3,696 sentences and a test set of 1,344 sentences. These sentences are meticulously transcribed at both the word and phonetic levels, providing valuable data for tasks such as acoustic-phonetic research, speech recognition, and speaker identification. TIMIT includes time-aligned orthographic and phonetic transcriptions and was recorded at a 16 kHz sampling rate [6], [45].
- **Harvard sentences:** The Harvard sentences, also known as Harvard lines, consist of a carefully curated set of 720 phonetically balanced phrases designed for standardized testing in telecommunications, including voice over IP, cellular networks, and telephone systems. These sentences were specifically designed to contain phonemes at frequencies that reflect their natural occurrence in English speech. Each sentence is spoken by both a male and a female speaker, ensuring a balanced representation of gender-based phonetic variations. According to [46], the collection is organized into 72 lists of 10 sentences each, ensuring thorough phonetic coverage. With a focus on acoustic consistency, the Harvard sentences are frequently utilized in speech processing, telecommunications, and audio quality assessment due to their reliable structure, facilitating uniform and repeatable speech evaluations. These sentences are typically used in environments with a 16 kHz sampling rate, aligned with common practices in speech research and quality analysis [47], [48].
- **DEMAND:** The diverse environments multichannel acoustic noise database (DEMAND) database offers a comprehensive collection of multi-channel recordings capturing real-world acoustic noise across a variety of environments. Created by Thiemann et al. [49], this dataset features recordings made using a 16-channel microphone array at a sampling rate of 48 kHz. The database spans environments such as domestic settings, offices, public spaces, natural areas, streets, and various modes of transportation. DEMAND is categorized into six distinct environment types, with over 72 recordings from 18 locations, making it a valuable resource for researchers exploring spatial sound characteristics, noise sources, and acoustic patterns. It facilitates the development and testing of algorithms in real-world acoustic conditions, particularly in speech enhancement and environmental noise assessment.

All audio signals were standardized to a 4-second duration, converted to mono, and re-sampled at 16 kHz to maintain consistency across multiple datasets. These datasets were then split into training and validation sets. For the Harvard database, sentences from sets 1, 2, 3, and 4 formed the training set, while the validation set consisted of sentences from sets 5, 6, 7, and 8. In the TIMIT dataset, clean speech utterances used for training and testing were drawn from the eight dialect regions, comprising recordings from 22 male and 11 female speakers. Since the DEMAND dataset contains only environmental noises from outdoor and indoor places, noisy versions of the Harvard and TIMIT datasets were generated using DEMAND chosen from the 6 categories, 4 of which are *inside* and 2 of which are *open-air* with SNR values ranging from −5 dB to 10 dB in 5 dB increments. Evaluating the same DL-based CI method across different speech datasets ensures that the proposed approach is thoroughly tested on unseen data, validating its effectiveness and generalizability. This approach enhances the robustness and reliability of the method across diverse acoustic conditions.

The corrsponding electrodograms corresponding to the pristine utterances from HARVARD and TIMIT Sentences were stored in.mat files. Additionally, the.mat files contained electrodograms obtained from the traditional ACE stimulating strategy for the clean utterances with 1000 pulses per second (pps) per electrode, serving as targets for our model. The default settings of the ACE software, integrated within the CCi-Mobile software [38], were employed to transform input sounds into electrodograms.

2) METRICS

To evaluate the proposed methods, several metrics were employed to assess performance across multiple aspects, including signal processing, statistical analysis, and speech quality measures. Utilizing a diverse set of metrics not only provides a comprehensive evaluation of our approach but also enables a more robust comparison with a wide range of state-of-the-art methods, each of which may prioritize some metrics while neglecting others. These metrics are [35]:

- **Linear correlation coefficient (LCC):** In the realm of electrodograms, the assessment of relationships between denoised and unaltered signals was conducted by applying the LCC. The analysis extended to evaluating performance across diverse frequency spectra by performing computations specific to individual channels. The LCC, as denoted by Equation 8, is determined as:

$$LCC = \rho = \frac{\mathrm{Cov}(LGF_{\mathrm{clean}}, LGF_{\mathrm{denoised}})}{\sigma_{LGF_{\mathrm{clean}}}\sigma_{LGF_{\mathrm{denoised}}}} \tag{8}$$

  Here, LGF represents the signals within the electrodogram domain, cov(*clean*, *noise*) signifies the covariance between the clean and the noisy signal, and σ denotes the the standard deviations from denoised and pristine electrodogram signals. This metric enabled a nuanced exploration of signal correlations, providing insights into performance variations across distinct frequency bands within the electrodogram data.
- **SNRi:** In the electrodogram domain, the SNR improvement (SNRi) metric is employed to evaluate the performance on unprocessed signals (noisy electrodograms). This metric compares the SNR of the denoised signal relative to the clean signal against the SNR of the noisy signal relative to the clean signal. The SNRi is determined using Equation 9 [26]:

$$\mathrm{SNRi} = 20 \log \frac{\mathrm{LGF}_{\mathrm{noisy}} - \mathrm{LGF}_{\mathrm{clean}}}{\mathrm{LGF}_{\mathrm{denoised}} - \mathrm{LGF}_{\mathrm{clean}}} \tag{9}$$

- **WRS:** Is a metric used to evaluate the percentage of correctly identified words in a speech perception task [50]. WRS provides a quantitative measure of how well spoken words are understood by listeners in the presence of noise or other degradations. It is a vital tool in assessing speech intelligibility and is widely used in research and clinical settings to evaluate speech perception abilities. Mathematically, WRS can be calculated using the following formula:

$$WRS = \frac{N_c}{N_t} \times 100\% \tag{10}$$

  where: N_c is the number of correctly identified words. N_t is the total number of words presented. A connection between STOI and WRS can be established using a mapping function. The WRS, denoted as $f(d)$, is determined through a logistic function as stated in the equation 11:

$$f(d) = \frac{100}{1 + \exp(a.d + b)} \tag{11}$$

  Here, d represents the STOI score, while the parameters a and b are adjusted to the data using a nonlinear least-squares approach. As the logistic function preserves monotonicity, the relationship between STOI and WRS remains unaffected. Different mapping functions may be required for diverse speech datasets. However, to present speech intelligibility more clearly than the STOI score, this study directly utilized parameter values from the English IEEE database without data fitting. In this context, with $a = -17.4906$ and $b = 9.6921$, WRS was employed alongside the STOI score to evaluate speech comprehension effectively.
- **Short-time objective intelligibility (STOI):** Is a fundamental quantitative metric used to evaluate the intelligibility of speech signals, particularly in scenarios involving noisy or time-frequency weighted speech. The STOI score hinges on the correlation analysis between clean speech signals (x) and processed speech signals (y) in short-time, time-frequency regions, providing detailed insights into the efficacy of speech enhancement and separation algorithms [51]. It is worth noting that STOI is sometimes referred to as VSTOI to specify that the evaluated speech is vocoded before being assessed.

  Within each one-third octave band, an intermediate intelligibility measure is calculated to gauge the correlation between the clean and processed speech signals. This measure includes local normalization and the computation of the signal-to-distortion ratio (SDR) for each TF-unit, enabling the estimation of the correlation coefficient for each TF-region. The normalized and clipped TF-unit (Y) is determined to align with the energy of the clean speech signal. The correlation coefficient ($d_j(m)$) for each TF-region is obtained by analyzing the means of the clean and processed TF-units. The final STOI score is then derived by averaging these correlation coefficients across all one-third octave bands and frames as presented in equation 12:

$$\mathrm{STOI} = \frac{1}{JM}\sum_{j,m} d_j(m) \tag{12}$$

where J is the number of one-third octave bands, and M is the total number of frames.

In this work, the STOI is utilized to predict the speech intelligibility performance of the tested algorithms. To compute this measure, the generated electrodograms are resynthesized using a sine wave vocoder [52], this latter is designed to simulate CI processing by directly mapping CI electric pulses to acoustic pulses. It uses a temporal frame-based n-of-m selection method, commonly found in ACE processing strategy, to replicate speech information's temporal and spectral features in both CI and acoustic modes. This approach aims to ensure that the simulated output closely matches the auditory experience of CI users.

B. MODEL SETTING

The study delves into the pivotal role of kernel size within the convolutional blocks of the model, initially set at 3, a determination supported by findings in [26]. This choice, deemed optimal for the DeepACE model, serves as the baseline. The hyper-parameter settings used for training can be seen in Table 2. However, the research aims to push boundaries by exploring a range of configurations, thereby necessitating the adaptation of this parameter to refine and tailor the model to its full potential.

TABLE 2. **Hyper-parameters used for training the models.**

Description	Value
Number of filters in the autoencoder	64
Length of the filters	32
Number of channels in the bottleneck blocks	64
Number of channels in the skip-connections	32
Number of channels in the convolutional blocks	128
Kernel size in the convolutional blocks	3
Number of convolutional blocks in each repeat	8
Number of repeats	3

C. TRAINING AND VALIDATION PERFORMANCE

The training process is meticulously controlled through a combination of early stopping and learning rate reduction strategies to optimize performance and prevent overfitting. The training begins with an initial learning rate of 1×10^{-3} and is scheduled to run for a maximum of 200 epochs. However, early stopping is employed with a patience of 5 epochs to avoid unnecessary computation and ensure efficient training. If the validation loss does not improve after 5 consecutive epochs, the training will automatically terminate, preventing overfitting.

Additionally, the learning rate is dynamically adjusted using a reduction mechanism that multiplies the current learning rate by a factor of 0.8 if the validation loss stagnates for 3 consecutive epochs. This adaptive learning rate strategy ensures that the model continues to learn efficiently even as the training progresses. The best model during the training process is continuously saved based on the lowest validation loss, ensuring that the final model represents the optimal performance achieved during the training cycle. Together, these mechanisms ensure a balanced and effective training process, minimizing the risk of overfitting while maximizing model performance within a reasonable time-frame.

The model training objective function combines MSE and binary cross-entropy (BCE), aligning with the approach used and thoroughly justified in the baseline model study [26]. In this manuscript, this combination is referred to as the **loss** function.

1) OBTAINED LOSS WHEN VARYING KERNEL SIZES

Table 3 provides a comprehensive overview of the top-performing DL models employed across various strategies, including the baseline model. This summary encapsulates key metrics such as model size, training, and validation losses, offering a valuable resource for understanding the relative strengths and effectiveness of different DL approaches within the context of this study. It is obvious that the MHA with TCN combination shows improvement over the baseline across multiple kernel sizes, particularly with lower training and validation losses, highlighting the benefits of MHA in enhancing temporal feature extraction. When comparing MHA-TCN with Bi-GRU models (12×3, 16×2, and 64×2 configurations), the MHA-TCN with Bi-GRU 64×2 configuration consistently achieves the lowest losses (e.g., at kernel size 16, it shows 0.0400/0.0463, outperforming others). This indicates that combining MHA with Bi-GRU layers enhances sequential modeling and memory retention, leading to better performance. Additionally, the MHA-TCN with Bi-GRU 16×2 setup also provides competitive results, demonstrating a balanced trade-off between model complexity and performance. Comparing the best-performing configuration (MHA-TCN with Bi-GRU 64×2) in terms of train/validation loss to the baseline model, it significantly reduces both metrics (e.g., 0.0400/0.0463 vs. 0.0578/0.0672), confirming the superiority of MHA and Bi-GRU integration in boosting performance. The advantages of MHA lie in its attention mechanism for important features, while Bi-GRU provides better temporal dynamics modeling, making MHA with Bi-GRU an optimal strategy for this study.

It is worth noting that the results reveals that for MHA with TCN, the minimum validation loss is not aligned with the minimum training loss, indicating instability in its performance across different kernel sizes. For example, at kernel size 32, the validation loss reaches a low of 0.0573, but the corresponding training loss is 0.0475, which is not the lowest training loss observed. This mismatch between the minimum training and validation losses suggests potential

TABLE 3. Performance comparison between the baseline and proposed DL-based CI models in terms of model size, training loss, and validation loss.

	Baseline model [26]		MHA with TCN		MHA-TCN with Bi-GRU 12x3		MHA-TCN with Bi-GRU 16x2		MHA-TCN with Bi-GRU 64x2	
Kernel size	**Model Size**	**Train / Valid Loss**	**Model Size**	**Train / Valid Loss**	**Model Size**	**Train/ Valid Loss**	**Model Size**	**Train/ Valid Loss**	**Model Size**	**Train/ Valid Loss**
16	592k	0.0578 / 0.0672	610k	0.0543 / 0.0670	634k	0.0449 / 0.0521	640k	0.0493 / 0.0553	804k	0.0400 / 0.0463
32	642k	0.0664 / 0.0806	659k	0.0475 / **0.0573**	683k	0.0494 / 0.0585	689k	0.0492 / 0.0581	853k	0.0499 / 0.0565
48	690k	0.0466 / 0.0583	708k	0.0523 / 0.0616	733k	0.0563 / 0.0659	738k	0.0461 / 0.0526	902k	0.0457 / 0.0519
64	740k	0.0627 / 0.0719	758k	0.0519 / 0.0658	782k	0.0447 / 0.0524	788k	**0.0350 / 0.0446**	951k	0.0430 / 0.0500
80	789k	0.0486 / 0.0600	807k	0.0755 / 0.0851	831k	0.0474 / 0.0561	837k	0.0397 / 0.0484	1001k	**0.0374 / 0.0447**
96	838k	0.1301 / 0.1277	856k	0.0506 / 0.0621	880k	**0.0404 / 0.0494**	886k	0.0358 / 0.0449	1050k	0.0375 / 0.0457
112	887k	**0.0454 / 0.0576**	905k	**0.0471** / 0.0607	929k	0.0453 / 0.0552	935k	0.0530 / 0.0636	1099k	0.0422 / 0.0505
128	936k	0.0648 / 0.0777	954k	0.0488 / 0.0574	978k	0.0758 / 0.0844	984k	0.0410 / 0.0505	1148k	0.0377 / 0.0461

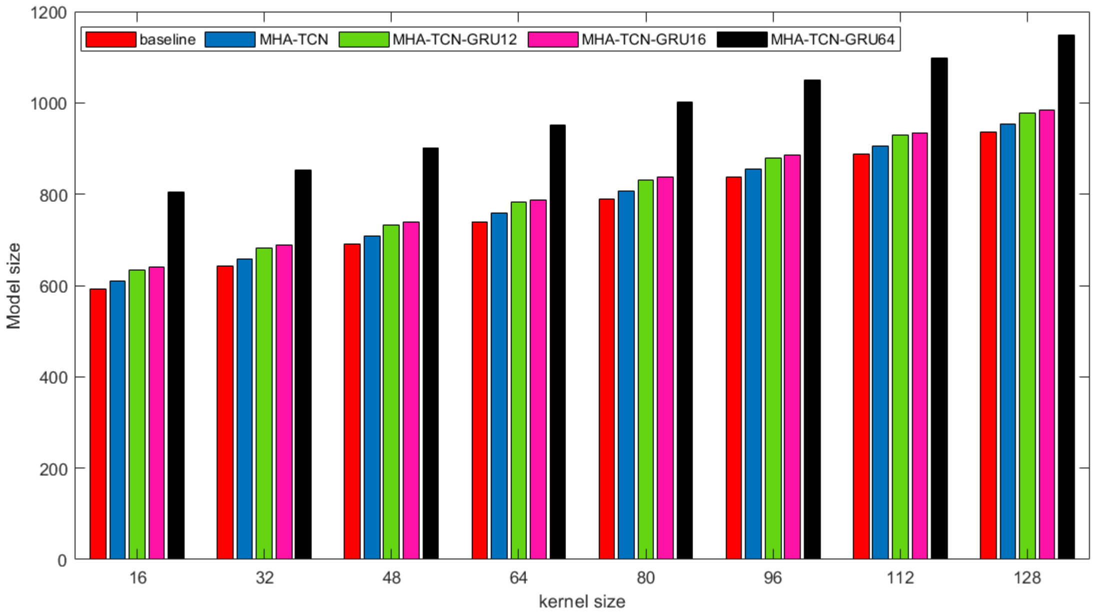


FIGURE 10. Comparison of model size at different kernel size values.

overfitting or underfitting issues, ranking MHA with TCN lower in terms of stability compared to models that show better consistency between these metrics.

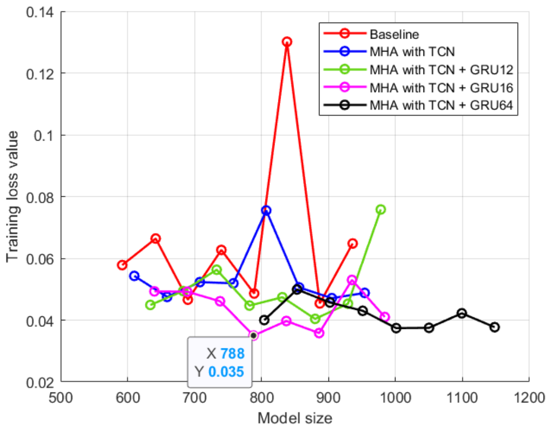
2) OBTAINED MODEL SIZE WHEN VARYING KERNEL SIZES

Figure 10 showcases a bar graph illustrating the impact of varying the kernel sizes on model performance. This visual aid clearly depicts how kernel size alterations influence the model's overall efficacy. It is obvious that MHA-TCN, MHA-TCN-GRU12, and MHA-TCN-GRU16 have closely aligned model sizes, with only slight increases between them. This suggests that adding GRU layers (as in GRU12 and GRU16) does not significantly inflate the model size, allowing for enhanced temporal feature modeling without a large jump in complexity. MHA-TCN-GRU64, however, shows a noticeable increase in model size across all kernel sizes. The larger size is a direct result of the deeper GRU layers (64 × 2), making it the most complex model in this comparison. When comparing MHA-TCN-GRU16 (the best among others) to the baseline, it offers a reasonable trade-off between model size and performance, likely outperforming the baseline while maintaining a manageable size, unlike the much larger MHA-TCN-GRU64.

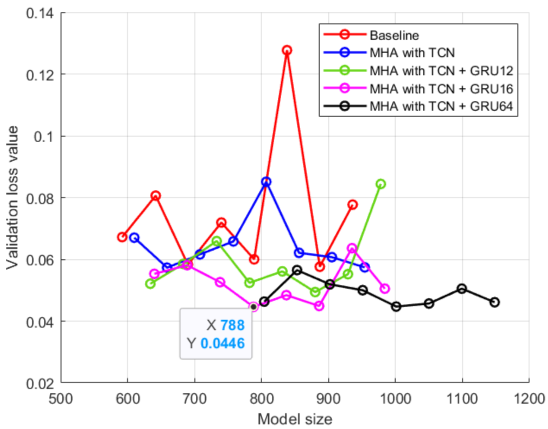
3) OBTAINED LOSS WHEN VARYING MODELS SIZES

Figure 11 illustrates the relationship between model size and both training and validation loss values across five different previously discussed model configurations.

In Figure 11 (a), the training loss is plotted as a function of model size for various strategies. The baseline model exhibits the highest and most erratic training loss values, with significant peaks around model sizes 800 and 900, indicating



(a) Training loss



(b) Validation loss

FIGURE 11. **Model size vs. training and validation loss values.**

poor performance and instability at these points. The MHA combined with TCN configuration achieves lower training losses compared to the baseline, though it still experiences a noticeable peak near model size 800 before stabilizing.

In contrast, the MHA combined with TCN and GRU models consistently achieve lower and more stable training losses. Among these, MHA with TCN + GRU12 shows a low and a stable training loss between model sizes 650 and 900. MHA with TCN + GRU16 performs slightly better in this range. For model sizes above 900, MHA with TCN + GRU64 demonstrates the best performance, though it cannot be trained effectively for model sizes below 800. The results were verified during the validation phase. Figure 11 (b) displays the validation loss as a function of model size, demonstrating a similar pattern with minor variations in the values.

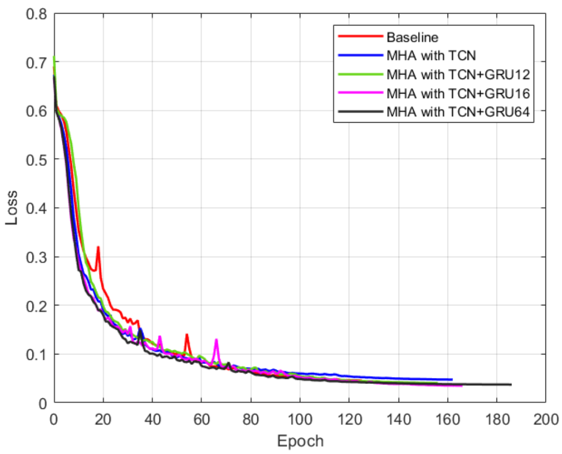
The selection of the optimal model for our research is guided by a balanced compromise between minimizing model size and achieving the lowest possible training and validation losses. After evaluating multiple model configurations, the MHA-TCN-BiGRU 16 $\times$ 2 model at a size of 788 emerged as the most suitable choice (Figure 11). This model demonstrates an optimal trade-off by offering one of the lowest validation loss values at 0.0446, which indicates strong generalization capabilities to unseen data. Furthermore, its training loss, recorded at 0.0350, is among the lowest across all models, showcasing its effective learning capacity without overfitting. Importantly, while some models, such as the MHA-TCN-BiGRU 64 $\times$ 2, achieved similarly low loss values, they required significantly larger model sizes (e.g., over 1000). In contrast, the MHA-TCN-BiGRU 16 $\times$ 2 achieves nearly equivalent performance while maintaining a relatively compact size. This balance makes it the ideal model, ensuring efficient computation and storage without compromising predictive accuracy, aligning well with our research objectives of building a performant yet efficient model for CI electrodogram prediction.

4) OBTAINED LOSS WHEN VARYING EPOCHS

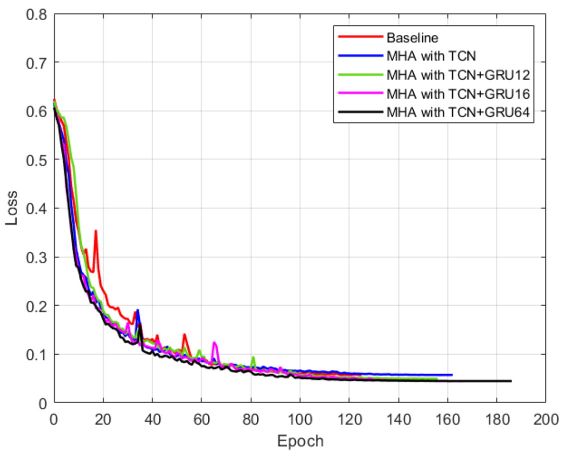
Figure 12 depicts the training loss curves for several models over 200 epochs. The aim is to evaluate and compare the training performance of these models. Initially, all models exhibit a rapid decline in training loss during the first 20 epochs. This sharp decrease indicates that the models effectively learn from the data at the beginning of the training process. However, as training progresses, the differences in the models' performances become more apparent. The baseline model consistently shows higher loss values throughout the training process compared to the other models. This suggests that the baseline model is less effective in learning from the data, resulting in inferior performance. On the other hand, the model that incorporates only MHA with TCN performs better than the baseline but still has slightly higher loss values than the models that also include GRUs. Models that combine MHA with TCN and various sizes of GRUs (12, 16, and 64 units) demonstrate superior performance. These models achieve lower loss values and exhibit better training convergence. Notably, while the MHA with TCN + GRU12 model shows some instability with occasional spikes in loss values during the first 100 epochs, it ultimately converges to a similar loss value as the other GRU-augmented models. This indicates that while there may be some variability during training, the overall learning capability of these models is enhanced by the inclusion of GRUs. The incorporation of MHA and TCN architectures, especially when combined with GRUs, significantly improves the training performance over the baseline model. These combinations result in lower loss values and more stable training processes. This analysis highlights the potential of these advanced architectures in enhancing model performance, suggesting that further optimization of GRU unit sizes and other hyperparameters could lead to even better results.

D. MODEL OPTIMIZATION THROUGH DATA AUGMENTATION

To enhance the efficacy of our best model, originally trained either on the Harvard Sentences dataset or TIMIT dataset, we embarked on a path of refinement by combining both



(a) Training curves



(b) Validation curves

FIGURE 12. **Comparative training and validation curves.**

datasets into our training regimen. This strategic amalgamation of datasets aimed to enrich the model's learning by exposing it to a broader spectrum of linguistic nuances and acoustic variations. Figure 13 illustrates the comparative performance of the baseline and best-performing model (TCN-MHA-BiGRU16) in terms of loss across 200 epochs, with training and validation curves clearly outlined.

The baseline model's training curve shows a steady decrease in loss, starting at above 0.5 and progressively lowering, demonstrating the model's learning process. However, the validation loss sometimes consistently higher than the training loss throughout the epochs, which points to potential overfitting, where the model may be too tailored to the training data and struggles to generalize to new, unseen data.

In contrast, the TCN-MHA-BiGRU16 model shows a more pronounced decline in both training and validation losses, starting at above 0.3 and progressively lowering. Within the first 20 epochs, the model exhibits a sharp drop in loss, which suggests fast adaptation and effective learning from the augmented dataset. This rapid decline stabilizes over time in epoch 80, indicating robust and fast learning

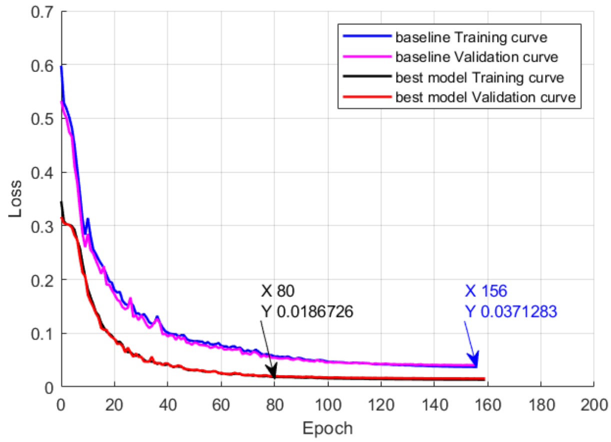


FIGURE 13. **Impact of data augmentation on the performance of TCN-MHA-BiGRU16 model and its comparison with the baseline model [26].**

without overfitting. The best model achieves a training loss of 0.0187 and a validation loss of 0.020, both considerably lower than those of the baseline model, which are 0.0371 for the training curve and 0.0407 for the validation curve in epoch 156.

The close alignment between the training and validation loss curves for the best-performing model demonstrates strong generalization capability, indicating its ability to perform consistently well on both datasets. This suggests that feeding the model a rich dataset, encompassing a broad spectrum of linguistic nuances and acoustic variations, significantly enhanced its accuracy and improved its ability to generalize effectively across diverse data.

E. OBTAINED ELECTRODOGRAMS AND SYNTHESIZED VOCODED SIGNALS

The methodology involved a multi-faceted approach to comprehensively assess the impact of different scenarios on auditory signal processing. Initially, a specific phrase, "The birch canoe slid on the smooth planks." spoken by a male speaker from the first set of HARVARD Sentences, was chosen as a fundamental testing benchmark. To simulate real-world conditions, noise from the DEMAND database, specifically selected from PRESTO noise, was introduced to the clean utterances at 10 dB. This noise addition aimed to mimic environmental challenges commonly encountered in speech communication. Subsequently, electrodogram data was extracted using the different scenarios as illustrated in Figure 14, to capture the intricate details of the auditory signal. The proposed strategies, illustrated in subfigures (d-g) of Figure 14, show a remarkable resemblance to the clean electrodigram in subfigure (a). These strategies incorporate advanced DL models that enhance electrodigram clarity under noisy conditions. Specifically, subfigures (d) and (e), which feature the MHA-TCN architecture combined with Bi-GRU layers, offer superior noise reduction compared to the baseline models. The clarity in subfigure (g), which

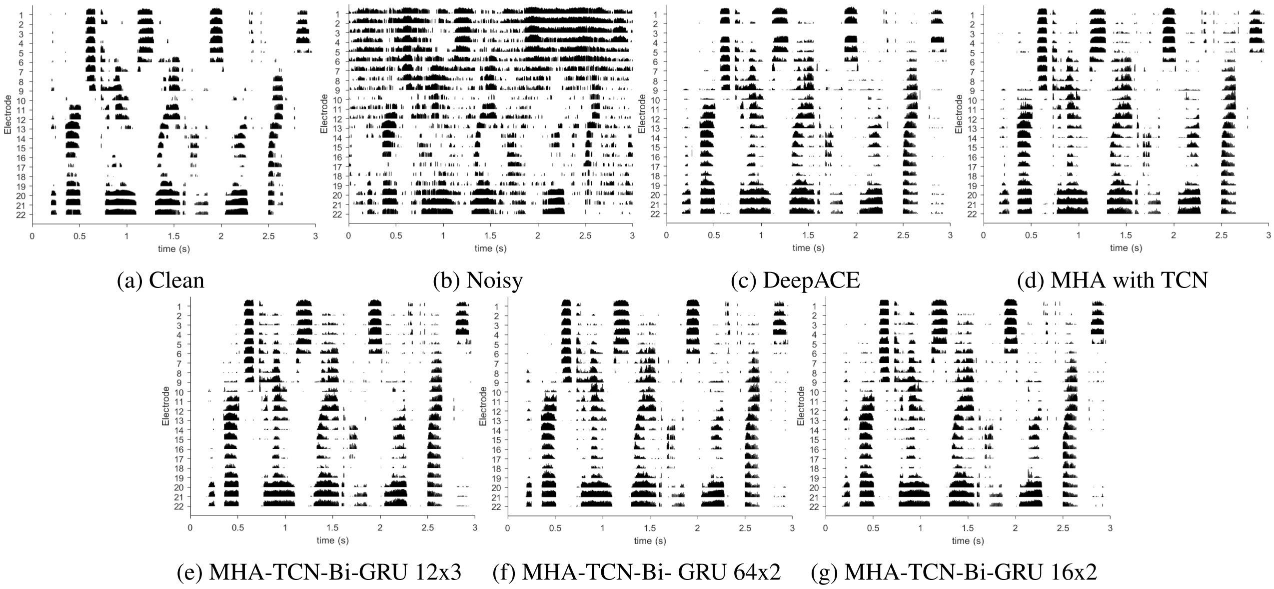


FIGURE 14. Electrodograms post-band selection and mapping for clean, noisy, DeepACE, and processed noisy speech signals under various proposed DL strategies.

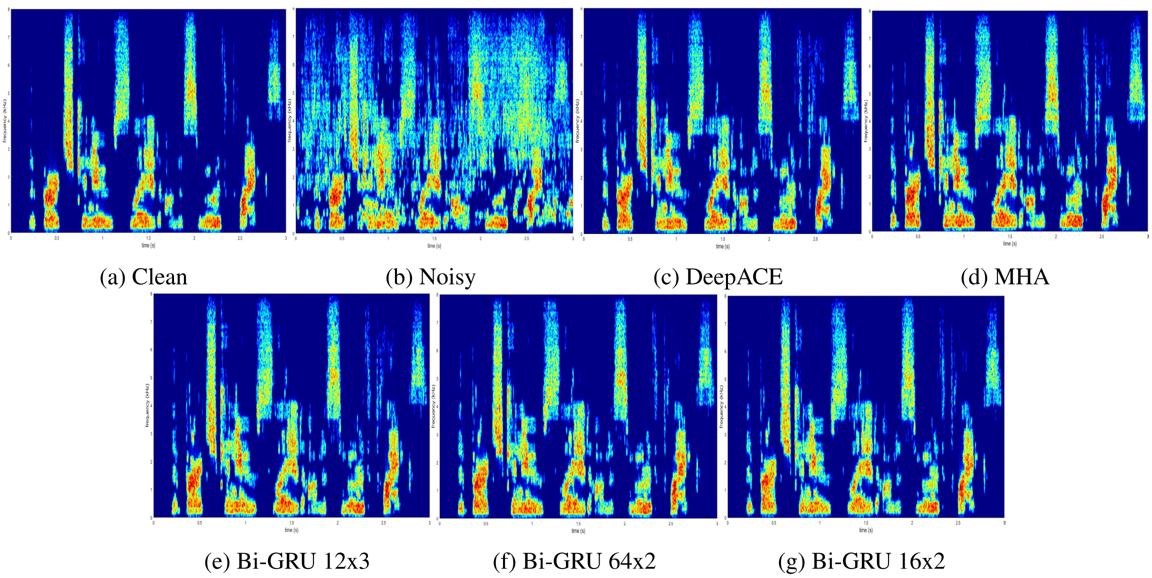


FIGURE 15. Spectrograms of vocoded signals derived from electrodograms for clean, noisy, DeepACE, and processed noisy speech signals using various proposed DL strategies.

uses the MHA-TCN model with Bi-GRU 16 × 2, closely approaches that of the clean reference, showcasing effective noise mitigation and preservation of fine speech signal details.

Figure 15 presents the spectrograms of the vocoded signals resynthesized from the electrodograms shown in the previous figure (Figure 14). These resynthesized signals, representing clean, noisy, and processed noisy speech under various proposed DL coding strategies, are used for objective evaluation with the STOI and WRS metrics. The figure highlights the effectiveness of each DL strategy in denoising the speech signals, illustrating how well each method restores the original audio quality and intelligibility. The figure provides insight into the comparison, demonstrating the impact of DL-based approaches on reducing noise and improving speech clarity in CI processing.

Table 4 presents the obtained LCC, STOI, and WRS values, which quantitatively characterize the similarity between the predicted electrodogram and the original version across diverse coding strategies. These metrics serve as a crucial indicator of the efficacy and accuracy of the prediction methods employed in this study. The baseline model shows

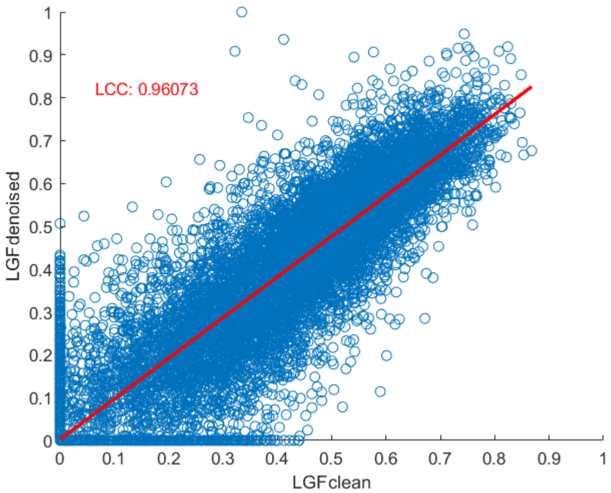


FIGURE 16. Correlation analysis between clean electrodogram and predicted electrodogram by TCN-MHA-BiGRU16 model.

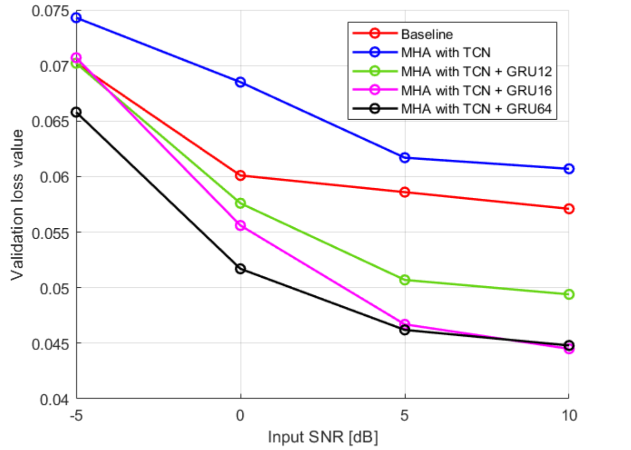


FIGURE 17. Validation loss vs. input SNR for different model architectures.

the lowest performance across all metrics. The MHA with TCN model provides a significant improvement over the baseline, with LCC increasing from 0.9235 to 0.9556, STOI from 0.8029 to 0.8205, and WRS from 98.7272 to 99.0612. The addition of Bi-GRU layers further enhances the results, with the MHA-TCN with Bi-GRU 16 $\times$ 2 achieving the highest performance. Overall, Bi-GRU variants outperform others.

The visualization depicted in the Figure 16 showcases the correlation between the clean electrodogram and the predicted electrodograms generated by the most optimal model (Bi-GRU with 16 units and 2 layers). This graphical representation offers a clear insight into the level of agreement or resemblance between the original electrodogram data and the predictions produced by the model deemed most effective for this specific analysis. By visually illustrating the correlation between the clean and predicted electrodograms, this figure provides a succinct yet comprehensive overview of the model's performance in capturing the underlying patterns and nuances within the electrodogram data.

F. COMPARATIVE ANALYSIS OF METRICS ACROSS MODELS UNDER VARYING SNR CONDITIONS

To assess the models' performance, tests were conducted using a set of mixed.wav files derived from the HARVARD sentences and TIMIT speech datasets, combined with environmental noises from the DEMAND dataset. The analysis compares the models' behaviors and performance trends, focusing on the effects of incorporating MHA and various BiGRU configurations on LCC values and STOI scores for vocoded signals.

1) ANALYSIS OF LOSS-SNR TRADE-OFF ACROSS MODEL ARCHITECTURES

Figure 17 illustrates the variation of validation loss against input SNR levels (-5, 0, 5, 10 dB) for different models: a baseline, MHA with TCN, and MHA with TCN combined with GRU layers (GRU12, GRU16, GRU64). The MHA with TCN model shows higher validation loss values than the baseline across all SNR levels, indicating poorer performance. Among the GRU variations, GRU64 achieves the lowest loss at every SNR level, significantly outperforming both the baseline and MHA with TCN. At -5 dB, GRU64 reduces the validation loss to approximately 0.045, compared to the baseline's 0.07 and MHA with TCN's 0.075. GRU16 slightly outperforms GRU12, demonstrating the advantage of increased capacity, but both GRU models provide considerable improvements over the baseline. Overall, integrating GRU layers with MHA and TCN significantly enhances performance, particularly under low-SNR conditions, while MHA with TCN alone fails to outperform the baseline.

2) COMPARATIVE ANALYSIS OF SNRi AND MODEL STABILITY

The comparative analysis evaluated five models—Baseline, TCN-MHA, TCN-MHA-BiGRU12, TCN-MHA-BiGRU16, and TCN-MHA-BiGRU64—using boxplots to assess SNRi across input levels of -5, 0, 5, and 10 dB. This analysis provides insights into both the magnitude of improvement and the stability of each model under varying noise conditions, as presented in Figure 18. The Baseline model showed consistent but tapering performance at higher SNRi levels, with moderate variability. TCN-MHA achieved robust improvements and stability across all conditions, benefiting from temporal convolutional mechanisms. TCN-MHA-BiGRU12 improved SNR but showed moderate variability and occasional outliers. TCN-MHA-BiGRU16 balanced performance and stability, offering consistent results. TCN-MHA-BiGRU64 demonstrated the highest SNRi and superior stability, excelling at higher SNRis, making it the most reliable model. This analysis underscores trade-offs, with TCN-MHA as an efficient option and TCN-MHA-BiGRU64 as the optimal choice for robust noise reduction.

TABLE 4. Comparison of LCCs for predicted electrodograms across various DL strategies.

Model variant	Baseline [26]	MHA with TCN	MHA-TCN with Bi-GRU 64x2	MHA-TCN with Bi-GRU 16x2	MHA-TCN with Bi-GRU 12x3
LCC	0.9235	0.9556	0.9560	0.9607	0.9590
STOI	0.8029	0.8205	0.8248	0.8345	0.8277
WRS	98.7272	99.0612	99.1287	99.2636	99.1714

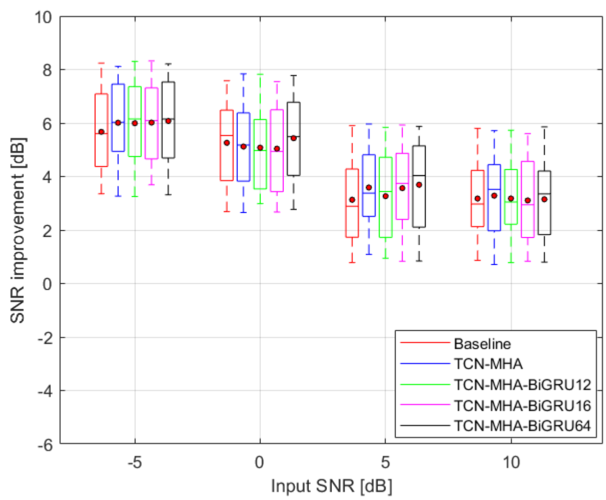
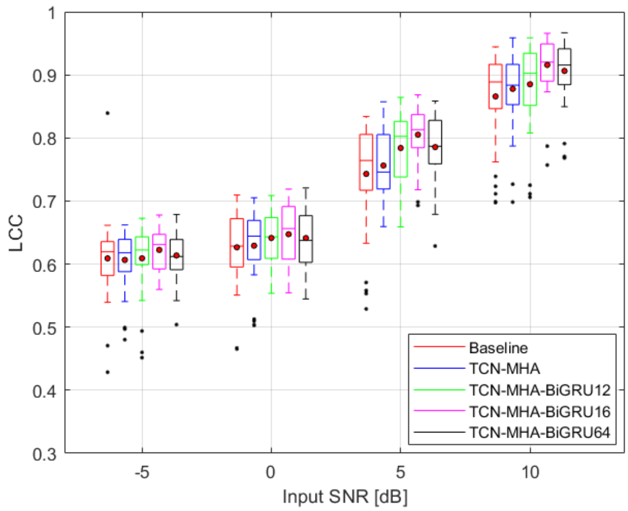
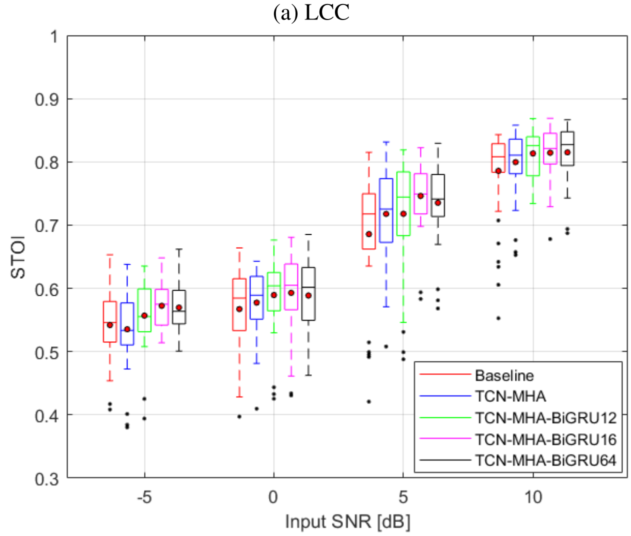


FIGURE 18. Box plots of SNRi in dB for the tested models. The horizontal lines within each box denote the median for each condition, while the circular markers represent the mean improvement. The top and bottom edges of the boxes correspond to the 3^{rd} percentile (Q3) and 1^{st} percentile (Q1), respectively. The length of each box reflects the interquartile range (IQR), which is used to define the whiskers that illustrate the variability beyond the upper and lower quartiles (the upper whisker is determined by $Q3 + 1.5 \cdot \text{IQR}$, and the lower whisker is determined by $Q1 - 1.5 \cdot \text{IQR}$). Black dots represent data points that fall outside the whisker range (outliers).



(a) LCC



(b) STOI

FIGURE 19. Comparative analysis of LCC and STOI performance. All pairwise comparisons revealed statistically significant differences.

3) COMPARISON IN TERMS OF LCC

The analysis of the LCC across the baseline, and different proposed DL models at varying SNR highlights the performance variations and the effectiveness of incorporating BiGRU layers. Figure 19 (a) illustrates the LCC obtained by the evaluated algorithms in different speech and noise conditions.

- At SNR of -5 dB, the baseline model exhibits a moderate degree of correlation with an LCC range from 0.4286 to 0.8393 and a median of 0.6196. Although it can maintain some correlation, its performance is inconsistent, particularly under challenging noise conditions. In contrast, the TCN-MHA model demonstrates slightly improved stability with a narrower LCC range of 0.4803 to 0.6621 and a median of 0.6178, suggesting a more reliable correlation between input and output signals. Introducing BiGRU layers significantly enhances performance, especially with TCN-MHA-BiGRU12, which shows a range from 0.4516 to 0.6727 and a median of 0.6225, indicating improved consistency in maintaining correlation. The TCN-MHA-BiGRU16 model stands out at this SNR level, exhibiting an LCC range from 0.5598 to 0.6779 with a median of 0.6310, showcasing its superior stability compared to the other models. Its higher minimum value suggests better correlation preservation even in the noisiest conditions. Although TCN-MHA-BiGRU64 achieves the highest maximum value at 0.6786, its broader range indicates more variability in performance. Overall, the LCC analysis at -5 dB SNR highlights that the BiGRU-enhanced models, particularly

TCN-MHA-BiGRU16, are more resilient in preserving signal integrity amidst significant noise interference.

- At the SNR of 0 dB, the baseline model's performance shows a range from 0.4655 to 0.7095, with a median of 0.6285, reflecting some improvement over the -5 dB condition. However, its variability remains evident. The TCN-MHA model again outperforms the baseline, with a narrower LCC range from 0.5029 to 0.7052 and a higher median of 0.6443. This improvement in consistency is further amplified in the BiGRU models. TCN-MHA-BiGRU16 achieves the highest median of 0.6563, along with a range from 0.5546 to 0.7190, underscoring its ability to maintain strong correlation even amid moderate noise. At this SNR, TCN-MHA-BiGRU64 presents a slightly lower median of 0.6376, but its peak score of 0.7208 indicates significant potential in optimal conditions. The overall trend at 0 dB SNR reaffirms the advantage of BiGRU integration, with TCN-MHA-BiGRU16 consistently demonstrating the best performance across various metrics, reinforcing its status as the most reliable model in moderate noise environments.
- As the noise conditions improve to SNR of 5 dB, the performance of all models is enhanced significantly. The baseline model shows a wider range from 0.5288 to 0.8339 with a median of 0.7643, suggesting improved stability. The TCN-MHA model further refines this performance with an LCC range of 0.6594 to 0.8573 and a median of 0.7459, indicating its robustness against noise. The BiGRU models, particularly TCN-MHA-BiGRU16, continue to excel with the highest median of 0.8132 and a range from 0.6931 to 0.8683. This model's strong performance across metrics reinforces its effectiveness in achieving high correlation preservation. At this level of SNR, TCN-MHA-BiGRU64, while presenting the lowest minimum among the BiGRU models, maintains a robust median of 0.7867. The peak performance of TCN-MHA-BiGRU16 signifies its capability for superior correlation fidelity. The improved median values across the board illustrate the effectiveness of the BiGRU layers in enhancing correlation maintenance. TCN-MHA-BiGRU16's consistent performance further solidifies its standing as the best model under moderately noisy conditions.
- At the SNR of 10 dB, the LCC analysis continues to demonstrate the strengths of the models. The baseline model maintains an LCC range from 0.6976 to 0.9444, with a median of 0.8888, showcasing decent consistency despite some dips at lower scores. The TCN-MHA model exhibits similar performance but with slightly higher upper-bound capabilities, reflected in its range of 0.6983 to 0.9586 and a median of 0.8836. However, the BiGRU models maintain the strongest correlation consistency, with TCN-MHA-BiGRU16 producing the highest median and peak performance.

Overall, the LCC analyses across different SNR levels reveal that TCN-MHA-BiGRU16 consistently outperforms the other models, particularly in challenging noise conditions. It exhibits a remarkable ability to maintain correlation, proving to be the most effective model in preserving the linear relationship between original and processed signals. This emphasizes the advantage of integrating BiGRU layers into the TCN-MHA framework, making TCN-MHA-BiGRU16 a suitable choice for real-world applications where noise interference is a concern.

4) COMPARISON IN TERMS OF STOI

The boxplot analysis of STOI scores across different SNR levels for the five models presented in Figure 19(b) reveals significant insights into their performance under varying noise conditions.

- At -5 dB SNR, the baseline model exhibits moderate intelligibility, with STOI scores ranging from 0.4084 to 0.6530. This range reflects its stable yet limited capability in challenging noise environments. The TCN-MHA model shows a slight improvement, with scores between 0.3802 and 0.6379, indicating enhanced robustness despite some variability. The TCN-MHA-BiGRU12 model presents a broader range of STOI values (0.3942 to 0.6353), suggesting a balanced performance but highlighting the potential for further enhancement. In contrast, the TCN-MHA-BiGRU16 model shows significant variability, with scores spanning 0.5141 to 0.6481, implying that it can achieve high intelligibility in some conditions, yet its performance is inconsistent. Notably, the TCN-MHA-BiGRU64 model stands out, with scores ranging from 0.5008 to 0.6618, demonstrating consistently higher intelligibility and improved robustness under low-SNR conditions. This trend suggests that the integration of BiGRU layers enhances performance, particularly in adverse noise environments.
- At SNR of 0 dB, performance differences among the models become more pronounced. The baseline model maintains consistent STOI values ranging from 0.3972 to 0.6639, indicating moderate intelligibility in noisy settings. The TCN-MHA model outperforms the baseline, with scores primarily between 0.4098 and 0.6429, reflecting a positive influence of the attention mechanism on speech clarity. The TCN-MHA-BiGRU12 model demonstrates a wider performance range (0.4254 to 0.6763), showcasing potential improvements in clarity but also sensitivity to specific noise patterns. The TCN-MHA-BiGRU16 model displays robust performance, with scores from 0.4340 to 0.6807, consistently ranking above the baseline and TCN-MHA, which suggests the architecture's efficacy in managing noise variations. Lastly, the TCN-MHA-BiGRU64 model excels with scores ranging from 0.4625 to 0.6852, indicating its superior capability in enhancing speech intelligibility at this SNR level.
- When the SNR reaches 5 dB, the models reveal varying performance levels. The baseline model achieves scores between 0.4208 and 0.8150, reflecting moderate

TABLE 5. Comparison of model parameters and inference time.

	Baseline model [26]		MHA with TCN		MHA-TCN with Bi-GRU 12x3		MHA-TCN with Bi-GRU 16x2		MHA-TCN with Bi-GRU 64x2	
Kernel size	**NP**	**IT (ms)**	**NP**	**IT (ms)**	**NP**	**IT (ms)**	**NP**	**IT (ms)**	**NP**	**IT (ms)**
16	592540	897.47	610652	1107.37	634724	1822.98	640636	1620.87	804316	1804.87
32	641692	897.50	659804	1113.36	683876	1903.48	689788	1658.57	853468	1822.74
48	690844	909.22	708956	1129.51	733028	1927.20	738940	1664.82	902620	1824.79
64	739996	917.09	758108	1138.23	782180	1960.73	788092	1687.68	951772	1869.22
80	789148	936.59	807260	1162.24	831332	1971.61	837244	1707.31	1000924	1875.34
96	838300	954.58	856412	1192.12	880484	1991.75	886396	1711.85	1050076	1882.22
112	887452	989.25	905564	1201.84	929636	2114.39	935548	1724.41	1099228	1908.44
128	936604	1004.43	954716	1244.29	978788	2154.04	984700	1785.43	1148380	1917.78

Abbreviations: Number of Parameters (NP), Inference Time (IT).

variability. The TCN-MHA model scores from 0.5080 to 0.8314, indicating a slight performance enhancement. The TCN-MHA-BiGRU12 model shows STOI scores from 0.4879 to 0.8188, suggesting comparable performance to TCN-MHA with some fluctuations. The TCN-MHA-BiGRU16 model consistently ranks high with scores between 0.5833 and 0.8224, highlighting its effectiveness at this SNR level. Meanwhile, the TCN-MHA-BiGRU64 model exhibits scores ranging from 0.5682 to 0.8293, indicating slightly higher variability than TCN-MHA-BiGRU16 but without consistently maintaining high performance across all samples.

- At SNR of 10 dB, the models display relatively strong performance, with the baseline model scoring between 0.5527 and 0.8431, showcasing reasonable consistency. The TCN-MHA model ranges from 0.6527 to 0.8716, performing better than the baseline but exhibiting variability. The TCN-MHA-BiGRU12 model achieves scores from 0.7340 to 0.8592, demonstrating consistent higher performance compared to previous models. The TCN-MHA-BiGRU16 model, with scores between 0.6779 and 0.8860, ranks among the top performers, indicating effective maintenance of intelligibility. Lastly, the TCN-MHA-BiGRU64 model shows scores ranging from 0.6938 to 0.8843, suggesting it performs well overall while exhibiting slightly more variability.

The TCN-MHA-BiGRU16 and TCN-MHA-BiGRU64 models emerge as the most effective configurations across SNR levels, with TCN-MHA-BiGRU16 providing the most consistent and highest intelligibility scores. This analysis underscores the importance of integrating BiGRU units, particularly with 16 and 64 units, to significantly enhance model performance in terms of speech intelligibility under varying noise conditions.

5) PARAMETER COUNT VS. INFERENCE TIME REQUIREMENTS

According to Table 5, the baseline model exhibits the smallest number of parameters (NP), with 592,540 at a kernel size of 16, achieving an inference time (IT) of 897.47 ms. As more complex architectures, such as MHA with TCN and Bi-GRU variants, are introduced, both NP and IT increase due to the added computational complexity. Increasing the kernel size further amplifies NP and IT across all models. For example, in the "MHA-TCN with Bi-GRU 64 $\times$ 2" configuration, NP grows from 804,316 (kernel size 16) to 1,148,380 (kernel size 128), while IT rises from 1,804.87 ms to 1,917.78 ms. Although larger models may offer better performance, the increased IT poses challenges for real-time applications. IT for most models exceeds 1,000 ms for kernel sizes above 48, making real-time deployment difficult. The inference time was measured on a machine equipped with an Intel Core i7 processor, 12GB RAM, and an Nvidia GeForce 940M GPU.

For real-time CI systems, maintaining a total latency under 30 ms is critical. Given the current inference times, significant optimization techniques such as pruning, quantization, or model distillation would be necessary to adapt larger models for real-time use. Simpler configurations, such as the baseline model or MHA with TCN for smaller kernel sizes (16 and 32), show promise for real-time deployment after further optimization.

In addition to all previous analysis, Table 6 offers a comprehensive evaluation of several models tested in our study alongside state-of-the-art models. This evaluation includes three models: ACE [26], Conv-TasNET+ACE [26], and DeepACE [26], all of which have been tested using the same STOI metric. The remaining columns in the table include the baseline and our four tested models (TCN-MHA, TCN-MHA-BiGRU12, TCN-MHA-BiGRU16, and TCN-MHA-BiGRU64). In this context, it is essential to note that the STOI scores of the reference models correspond to the mean values of their respective test datasets, whereas our presented values correspond to the mean values in noisy conditions.

Our study investigates performance across four different SNR levels: -5 dB, 0 dB, 5 dB, and 10 dB, with the results structured from the first to the fourth row, respectively. A key observation is that the STOI scores show a general

TABLE 6. Comparison of STOI scores and CCR/ WRS / SR metrics at different SNR levels across various state-of-the-art models and our proposed method.

Model	ACE [26]	Wiener + ACE [26]	Conv-TasNET + ACE [26]	DeepACE [26]	SNR Level	Electro-deNet -CS [29]	FCN [20]	MMSE [20]	DDAE [20]	Baseline [26]	TCN-MHA	TCN-MHA-BiGRU12	TCN-MHA-BiGRU16	TCN-MHA-Bi GRU64
STOI	0.807	—	0.789	0.803	-5 dB	—	—	—	—	0.5420 0.5418(*) 0.5421(**)	0.5354 0.5401(*) 0.5405(**)	0.5570 0.5565(*) 0.5568(**)	0.5725 0.5722(*) 0.5725(**)	0.5697 0.5700(*) 0.5697(**)
					0 dB	0.5899	—	—	—	0.5673 0.5678(*) 0.5782(**)	0.5775 0.5780(*) 0.5925(**)	0.5895 0.5923(*) 0.6042(**)	0.5929 0.5932(*) 0.6111(**)	0.5887 0.5930(*) 0.6094(**)
					1 dB	—	0.73(*), 0.77(**)	0.54(*), 0.55(**)	0.71(*), 0.75(**)	0.5932 0.6544(*) 0.6692(**)	0.6062 0.6602(*) 0.6881(**)	0.6173 0.6988(*) 0.7212(**)	0.6199 0.7261(*) 0.7771(**)	0.6148 0.7292(*) 0.7697(**)
					5 dB	—	0.79(*), 0.81(**)	0.67(*), 0.67(**)	0.78(*), 0.80(**)	0.6858 0.7689(*) 0.7987(**)	0.7176 0.7863(*) 0.8095(**)	0.7178 0.7768(*) 0.8099(**)	0.7460 0.8065(*) 0.8255(**)	0.7353 0.8082(*) 0.8206(**)
					10 dB	0.7213	—	—	—	0.7857 0.7725(*) 0.8098(**)	0.7995 0.7955(*) 0.8235(**)	0.8084 0.7906(*) 0.8342(**)	0.8145 0.8118(*) 0.8412(**)	0.8132 0.8255(*) 0.8419(**)
WRS/CCR/SR (%)	40.3* 42.9**	54.9*, 56.2**	54.7*, 54.3**	63.1*, 64.1**	-5 dB	28.8	—	—	—	44.71 44.63(*) 44.76(**)	41.88 43.89(*) 44.07(**)	51.25 51.04(*) 51.17(**)	57.96 57.84(*) 57.96(**)	56.77 56.89(*) 56.77(**)
					0 dB	74.2	—	—	—	55.73 55.95(*) 60.37(**)	60.08 60.29(*) 66.17(**)	64.99 66.10(*) 70.59(**)	66.33 66.45(*) 73.04(**)	64.67 66.37(*) 72.45(**)
					1 dB	—	76.9(*), 80.6(**)	65.2(*), 68.9(**)	61.6(*), 72.0(**)	66.45 85.24(*) 88.21(**)	71.31 86.47(*) 91.24(**)	75.12 92.62(*) 94.89(**)	75.96 95.29(*) 98.02(**)	74.29 95.53(*) 97.75(**)
					5 dB	—	—	—	—	90.91 97.72(*) 98.63(**)	94.58 98.31(*) 98.86(**)	94.60 98.01(*) 98.87(**)	96.63 98.80(*) 99.14(**)	95.96 98.84(*) 99.06(**)
					10 dB	—	—	—	—	98.29 97.85(*) 98.87(**)	98.65 98.55(*) 99.11(**)	98.84 98.43(*) 99.26(**)	98.96 98.91(*) 99.34(**)	98.93 99.14(*) 99.35(**)

* :SSN, ** :ICRA7 noise, (*): Engine noise, (**): Street noise.

improvement as the SNR conditions become more favorable. Specifically, the best STOI scores are achieved at 10 dB SNR, highlighted in the fourth row of Table 6. These scores not only emphasize the robustness of our tested models under various noise conditions but also underscore their superior performance in denoising tasks compared to the reference models. In particular, the baseline model, represented by the DeepACE with a kernel size of 112, achieved the best training and validation loss. Nevertheless, the models such as TCN-MHA-BiGRU12, TCN-MHA-BiGRU16, and TCN-MHA-BiGRU64, demonstrate further enhancements in STOI scores in the same model size. For instance, at 10 dB SNR, our model TCN-MHA-BiGRU16 achieves a STOI of 0.8145, surpassing the mean STOI of DeepACE, which achieved 0.803, indicating an enhanced capability for signal denoising. Similarly, the traditional ACE and the hybrid TasNET+ACE models, both yielding lower STOI scores under noiseless conditions, illustrate the substantial benefits of our approaches in handling noisy environments effectively. This comparison illustrates the advantages of our proposed models in processing speech signals for CI users, particularly under noisy conditions. Using mean STOI values in our dataset highlights the stability and robustness of our models across varying noise levels, supporting their potential utility in real-world auditory scenarios for CI users.

In terms of character correct rate (CCR), WRS, and sentence recognition (SR), our proposed models demonstrate significant performance improvements across various SNR levels when compared to conventional methods, as summarized in Table 6. At the lowest SNR level, −5 dB, traditional ACE-based models, such as ACE baseline (40.3% for speech-shaped noise (SSN) noise and 42.9% for ICRA7 noise), and Wiener+ACE and Conv-TasNET+ACE approaches (around 54%), exhibit limited recognition capabilities due to substantial noise interference. In contrast, our TCN-MHA and TCN-MHA-BiGRU models achieve higher WRS values, reaching up to 57.96%, indicating enhanced character and word recognition capabilities in highly adverse conditions. At 0 dB, ElectrodeNet-CS achieves a WRS of 74.2%, with our models following closely, scoring between 60.08% and 66.33%, further demonstrating our models' competitiveness in moderate noise settings. As noise decreases at 1 dB, our TCN-MHA-BiGRU models achieve CCR and WRS values exceeding 75%, highlighting the robust character and word recognition capabilities, with performance comparable to or surpassing FCN and DDAE methods. At 5 dB, our models exhibit substantial improvements, with WRS scores exceeding 90% and CCR values approaching 95%, achieving high accuracy in character and word recognition, particularly notable in SR. At the highest SNR level of 10 dB, our TCN-MHA-Bi-GRU12 and TCN-MHA-BiGRU16 models reach near-perfect recognition, with CCR, WRS, and SR values reaching up to 98.96%. This high performance underscores the effectiveness of our models in denoising and processing signals for CI processing, where accurate recognition across characters, words, and sentences is crucial. The results indicate that while traditional ACE-based models and Conv-TasNET+ACE provide a reasonable baseline, our models significantly improve speech intelligibility and comprehension accuracy across all SNR levels, which is crucial for denoising applications in CI processing. This trend underscores the efficiency of our proposed models in preserving essential speech characteristics, enhancing recognition accuracy, and adapting to varied noise types and intensities in CIs.

VI. CONCLUSION

In conclusion, this study presented a DL-based approach aimed at enhancing CI speech perception in noisy environments. By employing innovative architectures such as TCN, MHA, and GRU, the proposed models effectively enhance electrodogram prediction and speech intelligibility. This integration of components captures both local and global dependencies in speech signals, significantly improving performance over baseline models. Experimental results confirm that combining MHA with TCN and Bi-GRU outperforms traditional strategies, achieving lower training and validation losses. The models were validated on diverse datasets, including TIMIT and Harvard sentences, demonstrating robust generalization to real-world scenarios. The study underscores substantial gains in speech intelligibility, particularly in challenging noise conditions, as indicated by improvements in STOI and LCC metrics. Overall, the proposed framework represents a promising advance in CI sound-coding strategies, with potential applications for enhancing the auditory experience of CI users in everyday settings.

Future work will include conducting subjective listening tests to validate the proposed method's performance in real-world scenarios. Implementing the model on embedded hardware systems could facilitate these tests by enabling real-time processing and low-latency performance. Such hardware implementation would make subjective testing more practical and efficient, while reducing the complexities associated with software-based simulations. In addition, continued optimization and testing on larger datasets could further improve these results, paving the way for more sophisticated real-time CI systems.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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DATA AVAILABILITY

No data is available for this study.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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